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Paleomagnetism and rotation constraints for the middle Miocene southwestern Nevada volcanic field

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Abstract. Middle Miocene rocks of the southwestern Nevada volcanic field (SWNVF) lie across the projection of the Walker Lane belt within the Basin and Range province and thus provide an interesting opportunity to test for late Cenozoic vertical-axis rotation. Paleomagnetic data from individual ash flow sheets document no significant relative vertical-axis rotation among localities within central SWNVF, an area of relatively low stratal tilts and widely spaced faults. A time-averaged mean paleomagnetic direction (D = 351.4°, I = 52.7°, α_{95} = 4.5°) calculated from data from numerous separate rock units suggests that the central SWNVF underwent minimal counterclockwise vertical-axis rotation (R = -7.1° ± 6.6°) with respect to the North American craton. No clockwise vertical-axis rotation is found to support projection of dextral faults of the Walker Lane beneath the central SWNVF. Clockwise rotation of variable magnitude is common at numerous sites from southern and western margins of the field. These clockwise rotations probably reflect dextral shear strain developed at the interface between the little extended central SWNVF block and more strongly extended areas to the south and southwest of the field. Negligible rotation of 11.45-Ma to 13.25-Ma tuffs relative to the central SWNVF was found at the southeast margin of the field where 90° clockwise rotation at the northwest termination of the Las Vegas Valley shear zone had been postulated. Any clockwise rotation in this area must predate 13.25 Ma, and thus dextral shear within this part of the Walker Lane belt was not synchronous or connected across the southern margin of the field. Small counterclockwise vertical-axis rotation relative to the craton, as found for the central SWNVF block, might be a regional feature in the western Great Basin.

Introduction

Middle Miocene rocks of the southwestern Nevada volcanic field (SWNVF) [*Byers et al.*, 1976, 1989] lie across the projection of the Walker Lane zone within the Basin and Range province (Figure 1) and thus provide an opportunity to constrain paleomagnetically whether the SWNVF underwent vertical-axis rotation during late Cenozoic extension within the southern Great Basin. Recent paleomagnetic studies have revealed areas affected by vertical-axis rotation of known or presumed Cenozoic age within the province [*Gillett and Van Alstine*, 1982; *Hudson and Geissman*, 1987, 1991; *Nelson and Jones*, 1987; *Hagstrum and Gans*, 1989; *Wells and Hillhouse*, 1989; *Li et al.*, 1990; *Rosenbaum et al.*, 1991; *Palmer et al.*, 1991]. The amount, areal extent, and age of such rotations are still incompletely constrained because published paleomagnetic studies pertain to small parts of the Great Basin, and because the geologic expression of vertical-axis rotations is commonly cryptic. Consequently, the structural mechanisms responsible for such rotations are not well documented. Cenozoic volcanic rocks that were deposited within the province afford many opportunities for further paleomagnetic studies to address these uncertainties.

Paleomagnetic data obtained from volcanic rocks are used in two ways to test for tectonic rotations. Remanent directions from a number of sites from a single widespread volcanic unit can be compared to assess relative rotations if the unit acquired its magnetization quickly with respect to paleosecular variation of the geomagnetic field [*Magill et al.*, 1982; *Wells and Hillhouse*, 1989]. Such comparisons can yield precise estimates of rotations because they circumvent uncertainties produced by directional dispersion due to paleosecular variation. Without independent information, however, it is not possible to determine which sites actually rotated with respect to the stable craton. Moreover, such rotation estimates may be strongly affected by any site-specific errors such as unit misidentification or inaccurate tilt correction. An alternative method to assess rotation is to calculate a mean remanence direction from a number of temporally discrete volcanic units from a locality to average the paleosecular variation. This time-averaged direction is then compared to an expected direction calculated from an appropriate cratonic paleopole. Rotation estimates of this type commonly are not as precise as relative rotations determined from individual units unless they incorporate data from a large number of volcanic units because they must average paleosecular variation. Rotations based on time-averaged directions, however, have the advantage of being "absolute" in sense and magnitude relative to the craton and of being less affected by random site-specific errors.

For this study, paleomagnetic data from the SWNVF are used to assess vertical-axis rotation using both of these methods. Emphasis is placed on data from the central, less deformed region of the volcanic field to calculate a time-averaged mean direction for comparison with an expected direction for the stable North American craton. This comparison indicates that the central 2400 km^2 of the SWNVF experienced only minor vertical-axis rotation during the past 15 m.y. Directions of remanent magnetizations from individual ash flow cooling units from this central area are subsequently used to assess tectonic rotations in the marginal, more deformed parts of the volcanic field.

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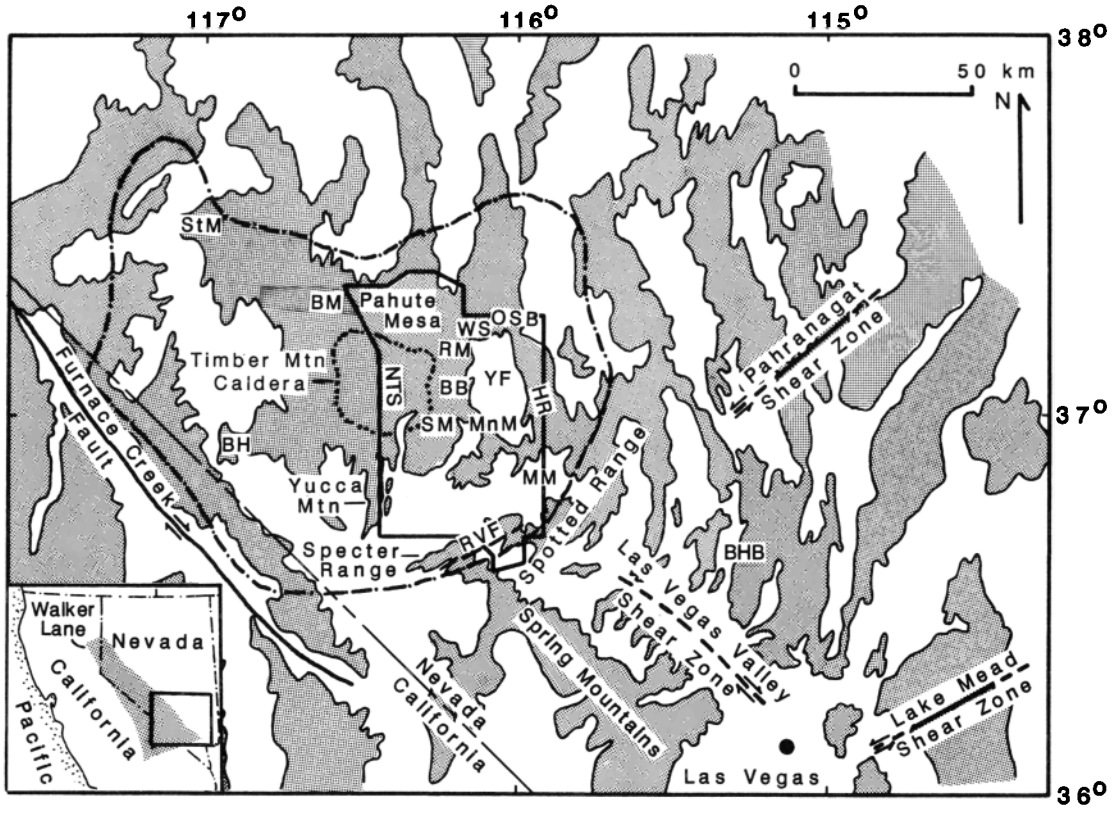


Figure 1. Generalized geological map of the southern Great Basin depicting upland (shaded) and alluvial (white) areas and selected structural features. Dash-dot line encompasses the distribution of middle Miocene rocks of the southwestern Nevada volcanic field. Shaded area in inset shows Walker Lane belt of *Stewart* [1988]. Abbreviations are BB, Big Butte; BH, Bullfrog Hills; BHB, Black Hills Basin; BM, Black Mountain; HR, Halfpint Range; MM, Massachusetts Mountain; MnM, Mine Mountain; NTS, Nevada Test Site; OSB, Oak Spring Butte; RM, Rainier Mesa; RVF, Rock Valley fault; SM, Shoshone Mountain; StM, Stonewall Mountain; and WS, Whiterock Spring.

Geologic Setting

The southern Great Basin (Figure 1) has been extending complexly during late Cenozoic time. High-strain zones that accommodated many tens of kilometers of extension [*Wernicke et al.*, 1988; *Hamilton*, 1988] are interspersed throughout the region. Upper crustal rocks in such zones commonly are tilted steeply and broken by closely spaced normal faults which may merge downward into regional detachment faults. Less strained crustal blocks such as the Spring Mountains lie between such high-strain zones. Extension in the southern Great Basin may have begun as early as Oligocene time [*Ekren et al.*, 1968; *Cemen et al.*, 1985; *Hamilton*, 1988], but most workers suggest that major extension began at about 16 Ma [*Anderson*, 1971; *Cemen et al.*, 1985; *Guth et al.*, 1988; *Wernicke et al.*, 1988; *Faulds et al.*, 1990; *Duebendorfer and Wallin*, 1991; *Fitzgerald et al.*, 1991]. The locus of strong extension generally migrated westward through time [*Wernicke et al.*, 1988].

Strike-slip faults were important components of the Cenozoic deformation of this part of the Basin and Range province [*Wright*, 1976]. Northeast striking sinistral faults and northwest striking dextral faults [*Tschantz and Pampeyan*, 1970; *Anderson*, 1973; *Longwell*, 1974; *Wright*, 1976; *Bohannon*, 1979] were active concurrently with Miocene extension. Northwest striking dextral faults lie within, and are considered to be an important structural component of, the Walker Lane belt [*Carr*, 1984; *Stewart*, 1988], a broad region of disrupted topography of the western Great Basin characterized by mixed normal and strike-slip faulting [*Stewart*, 1988]. Whereas *Stewart* [1988] concluded that Cenozoic dextral faulting was restricted to individual domains within the Walker Lane belt, previous workers have suggested that dextral shear was integrated and connected along the belt [*Albers*, 1967; *Carr*, 1984]. The SWNVF region lies across the northeastern margin of the southeastern Walker Lane belt (Figure 1).

Miocene rocks of the SWNVF were deposited on upper Precambrian and Paleozoic sedimentary rocks and, locally, on Mesozoic intrusive rocks. The SWNVF rocks have been studied in detail because of their use as a medium for underground nuclear testing at the Nevada Test Site and as a potential radioactive waste repository at Yucca Mountain [*Byers et al.*, 1976, 1989]. These rocks have been stratigraphically subdivided into many groups, formations, members, and informal units (Table 1). The volcanic rocks range in age from about 16 Ma to 7.5 Ma [*Kistler*, 1968; *Marvin et al.*, 1970; *Sawyer et al.*, 1990, in press] and comprise a sequence of dominantly rhyolitic and trachytic ash flow tuffs interlayered with lava flows and bedded

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tuffs [*Warren et al.*, 1989; *Broxton et al.*, 1989]. The Timber Mountain caldera complex is prominent in the center of the volcanic field. It retains topographic expression due to resurgent doming after eruption of the 11.45-Ma Ammonia Tanks Tuff [*Byers et al.*, 1976]. The Timber Mountain caldera complex was also the source of the 11.6-Ma Rainier Mesa Tuff. The younger Black Mountain caldera and Stonewall Mountain volcanic center to the northwest erupted the Thirsty Canyon Group (now dated at 9.4 Ma) and Stonewall Flat Tuff (now dated at 7.5 Ma), respectively [*Noble et al.*, 1984]. Calderas older than the Timber Mountain caldera are partly or completely buried by younger volcanic strata [*Orkild et al.*, 1968; *Byers et al.*, 1976; *Christiansen et al.*, 1977; *Sawyer and Sargent*, 1989]. Eruptive centers for many other volcanic units have not been confidently identified.

The central part of the SWNVF is least deformed by tectonic faulting and tilting that can be attributed to regional extension [*Orkild*, 1963; *Orkild et al.*, 1969; *Sargent and Orkild*, 1973; *Scott and Bonk*, 1984]. Volcanic strata in mesas around the Timber Mountain caldera complex (e.g., Pahute Mesa, Rainier Mesa, northern Shoshone Mountain, and northern Yucca Mountain) commonly dip gently ($\leq 10°$), and faults that offset them commonly are widely spaced and have less displacement than faults in peripheral areas of the field. A 70-km transect from Oak Spring Butte westward across Pahute Mesa to the Black Mountain area constitutes one of the longest east-west bedrock exposures uninterrupted by alluvial basins in the Great Basin. Volcanic rocks were strongly deformed in response to caldera resurgence in local areas of the central SWNVF such as at Timber Mountain.

Margins of the volcanic field were more extended than the central area. Beneath Yucca Flat and in the Halfpint Range in the eastern part of the field, volcanic strata commonly dip more steeply ($\geq 20°$) and are moderately extended along closely spaced normal faults [*Hinrichs and McKay*, 1965; *Byers and Barnes*, 1967; *Carr*, 1974, 1984]. In the Bullfrog Hills in the west part of the field, volcanic rocks are steeply tilted and intensely faulted where they overlie the detachment fault(s) separating them from Paleozoic and Precambrian rocks [*Maldonado*, 1990a, 1990b; *Maldonado and Hausback*, 1991].

Parts of the SWNVF were also affected by strike-slip deformation. For example, the east-northeast striking Rock Valley fault at the southern margin of the SWNVF has had sinistral offset [*Barnes et al.*, 1982]. *Burchfiel* [1965], *Ekren* [1968], and *Barnes et al.* [1982] attributed clockwise deflections of structures in Paleozoic rocks of the Specter Range and Spotted Range in the area of the Rock Valley fault to clockwise oroclinal rotation that accompanied movement of the dextral Las Vegas Valley shear zone [*Longwell*, 1974]. Clockwise rotation of the Tiva Canyon Tuff at southern Yucca Mountain [*Rosenbaum et al.*, 1991] might also be related to a buried dextral shear zone. A clockwise curvature of structures in SWNVF rocks of the Massachusetts Mountain area just north of the Specter and Spotted ranges, however, was shown by *Hudson* [1992] not to reflect clockwise vertical-axis rotation.

The extent to which dextral faults may affect the interior of the volcanic field is questionable. *Carr* [1984, 1990] considered the projection of the northeast boundary of the Walker Lane belt through the SWNVF area to be a sequence of right-stepping, northwest striking, dextral fault segments. In his model, dextral faults are found as far northeast as Pahute Mesa beneath the central part of the SWNVF. Carr further proposed that these dextral faults were active concurrently with development of the volcanic field and may have localized calderas in pull-apart structures at right steps between dextral faults. Throughgoing northwest striking dextral faults are not present at the surface through the central SWNVF, but lineament analysis [*Walker et*

Table 1 Stratigraphic Units of the Southwestern Nevada Volcanic Field

Unit	$^{40}Ar/^{39}Ar$ Age, Ma
Stonewall Flat Tuff	
Civet Cat Canyon Member	-
Spearhead Member	7.5
Thirsty Canyon Group	
Gold Flat Tuff	-
Trail Ridge Tuff	-
Pahute Mesa Tuff	-
Rocket Wash Tuff	9.4
comendite of Pillar Spring and comendite of Ribbon Cliff	-
Volcanics of Fortymile Canyon	
Beatty Wash Formation	-
basalt of Dome Mountain	-
Timber Mountain Group	
Ammonia Tanks Tuff	11.45
Rainier Mesa Tuff	11.6
rhyolite of Pinnacles Ridge	-
Paintbrush Group	
Tiva Canyon Tuff	12.7
Yucca Mountain Tuff	-
Pah Canyon Tuff	-
Topopah Spring Tuff	12.8
rhyolite of Delirium Canyon and rhyolite of Z	-
Calico Hills Formation	12.9
tuff of Blacktop Buttes	-
Wahmonie Formation	13.0
Crater Flat Group	
Prow Pass Tuff	13.1
Bullfrog Tuff (including Stockade Wash Tuff)	13.25
Tram Tuff	-
Belted Range Group	
Dead Horse Flat Formation	13.5
Grouse Canyon Tuff	13.7
comendite of Split Ridge	13.85
comendite of Quartet Dome	-
Lithic Ridge Tuff	14.0
Lava of Tram Ridge	14.0
Tub Spring Tuff	14.9
Older units	
tuff of Yucca Flat	15.1
Redrock Valley Tuff	15.25
Fraction Tuff	-
basalt of Whiterock Spring	-
tuff of Whiterock Spring	-

Stratigraphic nomenclature is from *Sawyer et al.* [in press], *Sargent and Orkild* [1973], *Christiansen and Noble*, [1968], *Hinrichs et al.*, [1967], and *Christiansen and Lipman*, [1965]. Informal stratigraphic units above "older units" are not placed in stratigraphic order relative to formal units. Isotopic ages are from *Sawyer et al.* [in press].

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al., 1986] and the truncation of subsurface formations [*Warren et al.*, 1985] suggest that northwest striking structures may underlie younger Timber Mountain and Thirsty Canyon strata beneath Pahute Mesa. *Minor* [1989] has noted zones containing small-scale, northwest striking dextral faults and conjugate sinistral faults in areas east of the Timber Mountain caldera. No evidence documents significant dextral offset on any of these known or inferred strike-slip zones, however.

Faulting occurred both during and after volcanism in the SWNVF. *Carr* [1984, 1990], *Warren et al.* [1985], and *Scott* [1990] give evidence for normal faulting that occurred prior to eruption of the 11.6-Ma Rainier Mesa Tuff in a north trending zone extending from Pahute Mesa to Yucca Mountain. *Minor* [1989] identified several phases of small-displacement faulting along the east side of the Timber Mountain caldera complex that spanned most of the field development. In other areas, however, major faulting and tilting appear to have entirely postdated eruption of the 11.45-Ma Ammonia Tanks Tuff [*Carr*, 1974, 1990; *Maldonado*, 1990a].

Methods and Data Characteristics

This study integrates newly acquired paleomagnetic data (96 sites) with recently published data (87 sites) from *Rosenbaum and Synder* [1985], *Rosenbaum et al.* [1991], and *Hudson* [1992]. Samples for this study were collected with a portable rock drill and oriented with a clinometer and both magnetic and solar compasses. At most sites, seven to nine samples were collected from outcrops spread over several tens of square meters. Sites suspected to lie within a basal flow of the Rainier Mesa Tuff of the Timber Mountain Group were not considered because directional dispersion has been recorded in this unit in some cases [*Hudson*, 1992]. For silicic lavas, only sites from flow interiors were used to avoid any dispersion related to post-blocking rotation of flow margins. Stratal attitudes of ash flow tuff sites were estimated by measurements of compaction foliations, stratigraphic contacts, and boundaries between welding zones, or were estimated from three-point calculations of extensive stratigraphic contacts. Attitudes for lava flows were derived from three-point calculations for extensive exposures or were extrapolated from nearby ashflow tuffs of similar age.

In the laboratory, samples were subjected to progressive alternating field or thermal demagnetization. At least six samples from each site were subjected to alternating field (AF) demagnetization (typically eight demagnetization steps per sample to peak inductions of 80 millitesla (mT)). Demagnetization results were plotted on vector diagrams and directions of components were determined for visually determined linear segments by principal component analysis [*Kirschvink*, 1980]. Most site means and their dispersion parameters (Table 2) were calculated using the statistics of *Fisher* [1953]. For a few sites affected by lightning strikes, site means were derived from intersecting remagnetization circles [*Halls*, 1976] using Bingham statistics [*Onstott*, 1980] and compared with mean directions derived using Fisher statistics on sample magnetizations for which the lightning-induced magnetization could be confidently removed. Mean directions derived using these two methods were typically very similar.

For this study we restricted our analysis to paleomagnetic data for which (1) the individual site mean directions were well defined and (2) paleohorizontal was well constrained. Additionally, care was taken to insure that an independent geomagnetic direction was recorded in each unit used in the calculation of a time-averaged mean direction. Multiple site means from individual ash flow tuff cooling units in the central SWNVF were averaged to derive a single unit mean for input into the calculation of the time-averaged direction and to serve as a reference direction for evaluation of tectonic rotations.

Rocks of the SWNVF commonly responded in an uncomplicated manner to progressive demagnetization (Figure 2). Most rock units carry one dominant, well-grouped component of magnetization that likely was acquired during initial cooling and is carried by one or a combination of the mineral phases magnetite, maghemite, and hematite. Detailed rock magnetic characteristics for several units have been given by *Rosenbaum* [1986, 1993], *Rosenbaum et al.* [1991], and *Hudson* [1992]. Lightning-induced isothermal remanent magnetizations have overprinted the characteristic magnetization at some sites. These spurious components are recognized by their high magnetic moments, low resistance to AF demagnetization, and dispersed directions. Data from samples for which lightning-induced overprints could not be convincingly removed were omitted from statistical calculations.

Rotation Constraints for the Central SWNVF

To evaluate possible vertical-axis rotation of the central SWNVF with respect to the craton, a time-averaged paleomagnetic direction for the volcanic field was compared to an expected direction derived from a middle Miocene North American paleopole. Paleomagnetic data were combined from several localities because no single structural block exposes all units of the SWNVF. Thus it is necessary to establish that these localities have not rotated relative to one another.

Comparisons Among Localities

Remanence directions carried by individual ash flow tuffs have been investigated to assess any relative rotations among sites from Pahute Mesa, Rainier Mesa, Aqueduct Mesa, Big Burn Valley, Oak Spring Butte, White Rock Spring, Big Buttes, Mine Mountain, Shoshone Mountain, and northern Yucca Mountain. Site mean directions from Trail Ridge, Ammonia Tanks, Rainier Mesa, Tiva Canyon, Bullfrog, Grouse Canyon, and Redrock Valley tuffs are shown in Figure 3. These units span much of the age of the volcanic field (Table 1). The southern Rainier Mesa locality exposes the greatest number (5) of these units in a single fault block. The tight clustering of site mean directions from each of these seven ash flow sheets indicates that no significant relative rotation occurred between the different sample localities.

Tuffs that were considered separately by *Byers et al.* [1976] as the Bullfrog Member of the Crater Flat Tuff at northern Yucca Mountain and as Stockade Wash Tuff at Big Butte and Rainier Mesa localities carry statistically identical remanence directions (Figure 3). In the revised stratigraphic sequence of *Sawyer et al.* [in press], these two units are considered to be correlative based on petrographic similarities and stratigraphic relations to older and younger units [*Warren*, 1983]. The concordance of remanence directions carried by these units suggests that they may have erupted closely enough in time to have

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TABLE 2 Paleomagnetic Data From the Southwestern Nevada Volcanic Field

Site	Lat°N	Long°W	N/N_o	k	α_{95}	Dg	Ig	St/Dp	Ds	Is	Locality	Rotation	Source
							Spearhead Member, Stonewall Flat Tuff						
JR89-19	37.296	116.582	7/8	1588	1.5	359.3	53.2	0/0	359.3	53.2	east of Black Mtn		1
							Comendite of Pillar Spring						
JR89-15	37.245	116.599	7/7	201	4.3	168.2	-43.5	170/4	164.4	-43.2	southeast of Black Mtn		1
91BH-32	37.243	116.645	8/8	532	2.4	165.4	-64.0	0/0	165.2	-64.0	south of Black Mtn		1
Average			2			164.8	-53.6						
							Trail Ridge Tuff						
JR89-5	37.211	116.458	5/7	816	2.7	170.3	-8.2	0/0	170.3	-8.2	southwest Pahute Mesa	1.5 ± 4.1	1
JR89-13	37.300	116.551	5/6	276	4.6	168.0	-12.8	315/2	168.4	-11.7	northeast of Black Mtn	-0.4 ± 5.1	1
JR89-18	37.238	116.596	3/6	741	4.5	167.6	-5.0	53/2	167.7	-6.8	southeast of Black Mtn	-1.1 ± 2.1	1
Average			3	807	4.3	168.8	-8.9						
							Rocket Wash Tuff						
JR89-4	37.210	116.456	7/7	788	2.2	143.5	-45.4	0/0	143.5	-43.4	southwest Pahute Mesa		1
91BH-27	37.251	116.699	8/8	309	2.8	144.4	-45.0	0/0	144.4	-45.0	west of Black Mtn		1
Average			2			143.5	-43.4						
							Comendite of Ribbon Cliff--Middle Lava						
JR89-16	37.243	116.599	6/6	507	3.0	13.8	57.2	40/2	16.7	58.0	southeast of Black Mtn		1
							Basalt of Dome Mountain						
90BH-9	36.977	116.440	8/8	423	2.7	1.3	51.1	180/2	358.8	51.1	west of Dome Mtn		1
							Ammonia Tanks Tuff						
7YF-3	37.174	116.208	9/9	256	3.2	2.6	62.6	180/4	354.9	62.5	Rainier Mesa	4.4 ± 7.2	2
8YF-29	37.223	116.167	5/9	239	5.0	342.2	55.9	205/2	349.6	57.0	Aqueduct Mesa	-0.9 ± 8.7	2
JR89-2	37.197	116.479	7/7	1118	1.8	355.7	64.5	220/2	352.9	63.1	southwest Pahute Mesa	2.4 ± 5.6	2
9YF-37*	36.995	116.226	8/8	---	2.9	354.0	63.0	205/2	350.8	61.9	Shoshone Mtn	0.3 ± 6.8	2
91YF-5	37.274	116.383	8/8	147	4.6	348.1	61.1	325/3	352.9	59.8	Pahute Mesa	2.4 ± 8.7	1
91YF-7	37.272	116.341	4/8	354	4.9	341.9	66.9	0/0	341.9	66.9	Pahute Mesa	-8.6 ± 11.0	1
91YF-9	37.268	116.322	8/8	442	2.6	349.5	63.1	0/0	349.5	63.1	Pahute Mesa	-1.0 ± 6.5	1
Average			7	510	2.7				350.5	62.1			
91YF-11	37.312	116.309	8/8	122	5.0	10.7	65.5	0/0	10.7	65.5	northeast Pahute Mesa	20.2 ± 10.8	1
8YF-10	36.959	115.932	8/8	202	3.9	36.4	60.5	170/38	326.5	65.1	Halfpint Range	-24.0 ± 8.8	2
91YF-23	36.989	115.875	8/9	630	2.1	327.4	67.9	266.5/9	335.2	59.7	Halfpint Range	-15.3 ± 5.7	1
7YF-15	36.926	116.032	7/7	393	3.1	2.8	47.1	135/12	351.7	55.2	CP Hogback	1.2 ± 6.3	2
7YF-18	36.861	115.992	4/7	432	4.4	345.4	43.7	45/12	353.1	53.7	Massachusetts Mtn	2.6 ± 7.5	2
JR87-16	36.895	115.988	4/8	375	4.8	298.4	40.8	340/24	321.1	53.3	Massachusetts Mtn	-29.4 ± 7.9	2
8YF-22	36.878	115.980	6/6	225	4.5	326.8	40.2	42/26	339.8	64.5	Massachusetts Mtn	-10.7 ± 9.6	2
8YF-24	36.896	115.980	7/7	479	2.8	332.8	24.4	45/33	345.0	54.4	Massachusetts Mtn	-5.5 ± 6.0	2
8YF-25	36.895	115.972	8/8	87	6.0	358.6	42.2	58/21	14.8	58.8	Massachusetts Mtn	24.3 ± 10.4	2
8YF-34	36.902	115.981	8/8	337	3.0	356.5	34.8	85/27	357.6	61.8	Massachusetts Mtn	7.1 ± 6.9	2
8YF-35	36.886	115.989	9/9	37	8.6	329.0	31.5	50/25	347.2	55.6	Massachusetts Mtn	-3.3 ± 13.1	2
JR87-5a	36.740	116.212	15/15	186	2.8	334.4	42.7	42/15	342.2	56.2	Skull Mtn	-8.3 ± 6.1	1
91YF-36	36.687	116.096	6/9	105	6.6	352.9	41.5	78/34	2.5	75.2	Spotted Range	12.0 ± 21.9	1
91YF-38	36.748	116.030	8/8	200	3.9	339.6	3.7	66/63	345.0	66.4	Pink Holes Hill	-5.5 ± 9.1	1
91YF-40	36.743	116.028	8/8	1431	1.5	325.1	49.4	22/27	0.4	67.5	Pink Holes Hill	9.9 ± 5.6	1
JR87-12	36.691	116.489	8/8	308	3.2	308.9	64.3	0/12	332.3	72.1	Yucca Mtn	-18.2 ± 9.6	1
8MH-7	36.710	116.579	9/9	889	1.5	346.8	65.2	330/8	1.5	61.9	Yucca Mtn	11.0 ± 5.3	1
8MH-10	36.715	116.553	9/9	103	5.1	16.2	57.9	275/2	15.6	55.9	Yucca Mtn	25.1 ± 8.6	1
91MH-1	36.695	116.494	8/8	458	2.6	326.8	59.3	350/6	336.8	61.2	Yucca Mtn	-13.7 ± 6.3	1
91MH-2	36.714	116.512	9/9	678	2.0	354.3	64.8	272/2	354.8	62.8	Yucca Mtn	4.3 ± 5.8	1
92MH-2	36.715	116.549	8/8	589	2.3	353.9	67.3	17/14	3.3	68.6	Yucca Mtn	12.8 ± 6.8	1
92MH-3	36.714	116.583	10/10	582	2.0	355.2	67.1	302/6	2.1	62.1	Yucca Mtn	11.6 ± 5.8	1
91BH-10	37.192	116.731	8/8	684	2.1	333.4	63.6	348/7	347.7	64.5	north of Sleeping Butte	-2.8 ± 6.1	1

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TABLE 2 (continued)

Site	Lat^{oN}	Long^{oW}	N/N$_o$	k	α_{95}	Dg	Ig	St/Dp	Ds	Is	Locality	Rotation	Source
							Rainier Mesa Tuff						
7YF-2	37.167	116.223	6/8	487	3.0	181.5	-49.0	163/6	170.1	-48.3	Rainier Mesa	-0.7 ± 4.2	2
7YF-4	37.182	116.221	7/7	650	2.4	173.6	-53.4	180/4	168.4	-52.8	Rainier Mesa	-2.4 ± 3.8	2
7YF-5	37.203	116.203	7/7	2475	1.2	169.9	-51.3	180/2	170.5	-51.5	Rainier Mesa	-0.3 ± 2.6	2
8YF-54	37.170	116.221	29/29	101	2.7	178.1	-47.6	142/9	168.9	-52.3	Rainier Mesa	-1.9 ± 4.1	2
7YF-16	37.128	116.171	20/25	194	2.4	161.7	-49.0	5/12	175.9	-52.4	Aqueduct Mesa	5.1 ± 3.8	2
9YF-36	36.995	116.225	7/7	745	2.2	172.6	-52.8	205/2	170.4	-51.7	Shoshone Mtn	-0.4 ± 3.6	2
91YF-3	37.282	116.415	7/8	757	2.2	174.4	-50.7	266/.5	174.4	-50.2	western Pahute Mesa	3.6 ± 3.5	1
91YF-4	37.273	116.385	8/8	512	2.5	168.3	-47.8	325/3	171.2	-46.5	Pahute Mesa	0.4 ± 3.6	1
91YF-8	37.272	116.339	8/8	839	1.9	167.9	-49.7	0/0	167.4	-49.7	Pahute Mesa	-3.4 ± 3.2	1
Average			9	874	1.7				170.8	-50.6			
91YF-6	37.357	116.377	8/8	272	3.4	171.7	-45.9	329/4	175.4	-44.2	northern Pahute Mesa	4.6 ± 4.4	1
91YF-10	37.352	116.317	8/8	488	2.5	177.8	-58.2	115/3	175.4	-60.8	northern Pahute Mesa	4.6 ± 4.6	1
91YF-29	37.321	116.249	8/9	152	4.5	214.4	-52.0	263/3	210.9	-50.8	Kawich Valley	40.1 ± 6.1	1
7YF-8	36.965	115.983	8/8	867	1.9	208.4	-54.6	170/21	175.5	-62.8	Halfpint Range	4.7 ± 4.0	2
7YF-10	36.984	115.981	10/10	573	2.0	193.1	-50.9	171/18	169.6	-54.4	Halfpint Range	-1.2 ± 3.5	2
7YF-12	37.119	116.007	5/7	1288	2.1	172.9	-43.2	165/23	151.4	-42.0	Halfpint Range	-19.4 ± 3.1	2
7YF-13	37.039	115.982	7/7	804	2.1	183.2	-57.8	170/5	175.2	-58.5	Halfpint Range	4.7 ± 3.9	2
8YF-9	36.959	115.931	7/7	625	2.4	208.2	-47.8	170/38	165.4	-55.2	Halfpint Range	-5.4 ± 4.0	2
9YF-25	37.015	115.971	8/8	383	2.8	197.4	-50.1	172/22	168.8	-54.5	Halfpint Range	-2.0 ± 4.4	2
9YF-26	37.014	115.971	8/8	1720	1.3	195.1	-43.0	172/22	172.9	-47.7	Halfpint Range	2.1 ± 2.6	2
9YF-39	36.974	115.987	8/8	1098	1.7	187.3	-54.4	184/17	164.4	-51.9	Halfpint Range	-6.4 ± 3.1	2
91YF-22	36.989	115.874	10/10	993	1.5	146.0	-59.6	266.5/9	152.1	-51.6	Halfpint Range	-18.7 ± 2.9	1
7YF-14	36.926	116.031	12/13	850	1.5	181.2	-42.9	135/12	171.5	-50.9	CP Hogback	0.7 ± 2.9	2
7YF-17	36.861	115.993	10/10	558	2.1	169.5	-42.7	45/12	177.7	-52.1	Massachusetts Mtn	6.9 ± 3.5	2
7YF-19	36.870	115.988	8/8	380	2.9	164.8	-28.0	30/24	178.5	-43.1	Massachusetts Mtn	7.7 ± 3.8	2
JR87-17	36.895	115.990	4/8	544	3.9	123.6	-36.3	340/26	145.3	-47.9	Massachusetts Mtn	-25.5 ± 5.1	2
8YF-12	36.894	115.937	6/9	181	5.0	172.0	-25.7	71/11	173.4	-36.5	Massachusetts Mtn	2.6 ± 5.4	2
8YF-14	36.903	115.941	5/7	1340	2.1	161.5	-30.7	110/4	157.0	-41.0	Massachusetts Mtn	-13.8 ± 3.1	2
8YF-16	36.913	115.948	5/7	547	3.3	147.1	-19.3	70/22	143.9	-40.6	Massachusetts Mtn	-26.9 ± 4.1	2
8YF-17	36.916	115.952	7/7	164	4.7	150.4	-40.8	37/10	154.8	-49.8	Massachusetts Mtn	-16.0 ± 6.2	2
8YF-18	36.914	115.956	8/8	684	2.1	170.9	-37.3	55/20	181.8	-54.5	Massachusetts Mtn	11.0 ± 3.6	2
8YF-21	36.877	115.979	8/8	409	2.7	167.3	-35.9	40/23	183.4	-52.3	Massachusetts Mtn	12.6 ± 4.1	2
8YF-23	36.894	115.982	7/8	986	1.9	138.7	-33.6	65/15	140.2	-48.5	Massachusetts Mtn	-30.6 ± 3.1	2
8YF-26	36.894	115.966	8/8	1471	1.4	179.1	-25.1	60/26	190.0	-46.7	Massachusetts Mtn	19.2 ± 2.7	2
8YF-31	36.912	115.992	7/8	154	4.9	190.8	-10.3	100/33	189.1	-45.2	Massachusetts Mtn	18.3 ± 6.0	2
8YF-33	36.903	115.981	9/10	137	4.4	183.0	-29.3	85/25	186.9	-53.9	Massachusetts Mtn	16.1 ± 6.4	2
8YF-36	36.886	115.990	6/8	265	4.1	140.2	-33.6	19/29	159.6	-56.8	Massachusetts Mtn	-11.2 ± 6.4	2
JR87-5r	36.740	116.212	8/8	276	3.3	163.8	-40.5	42/15	173.3	-52.6	Skull Mtn	2.5 ± 4.8	1
91YF-41	36.743	116.029	6/7	753	2.4	158.2	-44.1	22/27	188.2	-57.7	Pink Holes Hill	17.4 ± 4.2	1
90BH-234	36.953	116.658	23/23	390	1.5	188.9	-52.2	178/9	177.1	-53.0	Beatty Wash	6.3 ± 2.9	1
JR87-15	36.814	116.489	8/8	880	1.9	164.3	-40.9	5/18	180.8	-44.9	Yucca Mtn	10.0 ± 3.0	1
8MH-6	36.710	116.581	7/7	252	3.8	189.7	-58.1	330/8	197.5	-52.6	Yucca Mtn	26.7 ± 5.5	1
91BH-1	37.099	116.717	8/8	676	2.1	148.7	-44.2	95/15	136.7	-55.4	south of Sleeping Butte	-34.1 ± 3.7	1
91BH-8	37.188	116.728	8/8	272	3.4	198.8	-51.0	285/6	198.4	-45.0	north of Sleeping Butte	27.6 ± 4.4	1
91BH-17	37.306	116.809	8/8	992	1.8	145.4	-46.9	340/23	170.9	-47.7	northwest of Tolicha Peak	0.1 ± 3.0	1
91BH-18	37.307	116.805	8/8	839	1.9	150.6	-49.3	340/23	177.1	-47.7	northwest of Tolicha Peak	6.3 ± 3.1	1
91BH-14	37.295	116.964	8/8	198	4.0	161.6	-56.7	343/20	189.5	-52.2	east of Scottys Junction	18.7 ± 5.7	1
							Rhyolite of Pinnacles Ridge						
92YF-1	36.917	116.467	8/8	207	3.9	172.7	-37.5	62/12	177.1	-48.6	north of Yucca Mtn		1
							Tiva Canyon Tuff						
91YF-43	37.191	116.392	8/8	1523	1.4	164.6	-38.7	340/2	166.2	-38.5	southern Pahute Mesa	-1.6 ± 3.0	1
7YF-1	37.167	116.223	8/8	782	2.0	171.7	-36.3	163/6	167.3	-36.9	Rainier Mesa	-0.5 ± 3.3	3
8YF-44	37.961	116.218	8/8	718	2.1	171.3	-43.0	0/0	171.3	-43.0	Shoshone Mtn	3.5 ± 3.5	3
9YF-35	36.994	116.224	8/8	1256	1.6	168.6	-43.7	205/2	167.1	-42.5	Shoshone Mtn	-0.7 ± 3.1	2
JR87-4	36.902	116.487	11/11	829	1.6	163.9	-42.0	0/0	163.9	-42.0	northern Yucca Mtn	-3.9 ± 3.1	3

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TABLE 2 (continued)

Site	Lat°N	Long°W	N/N_o	k	α_{95}	Dg	Ig	St/Dp	Ds	Is	Locality	Rotation	Source
							Tiva Canyon Tuff (continued)						
JR81-2	36.882	116.461	9/9	420	2.5	166.3	-39.7	22/3	168.4	-41.4	northern Yucca Mtn	0.6 ± 3.7	3
M79-25	36.874	116.464	6/6	474	3.1	173.1	-39.6	76/5	173.7	-44.6	northern Yucca Mtn	5.9 ± 4.3	3
M79-26	36.871	116.463	5/7	302	4.4	164.2	-37.8	73/5	164.4	-42.7	northern Yucca Mtn	-3.4 ± 5.4	3
M79-28	36.870	116.459	6/6	625	2.7	165.0	-41.6	73/5	168.0	-46.6	northern Yucca Mtn	0.2 ± 4.1	3
Average			9	477	2.4				167.8	-42.1			
JR84-16	36.865	116.404	7/8	512	2.7	177.8	-38.3	69/8	180.3	-45.8	Yucca Mtn	12.5 ± 4.0	3
JR80-2	36.861	116.443	7/7	1124	1.8	164.3	-41.2	42/4	166.3	-44.5	Yucca Mtn	-1.5 ± 3.3	3
M79-7	36.851	116.447	5/6	323	4.3	169.7	-43.1	15/5	174.2	-45.1	Yucca Mtn	6.4 ± 5.5	3
JR80-1	36.844	116.452	7/7	365	3.2	167.9	-44.6	357/6	173.9	-45.2	Yucca Mtn	6.1 ± 4.5	3
JR84-15	36.842	116.408	6/6	1028	2.1	175.5	-40.5	11/9	184.5	-42.3	Yucca Mtn	16.7 ± 3.4	3
JR81-8	36.824	116.459	6/6	1863	1.6	173.1	-41.7	4/10	182.2	-42.8	Yucca Mtn	14.4 ± 3.1	3
JR84-13	36.811	116.415	8/10	599	2.3	164.5	-36.4	13/7	169.4	-39.5	Yucca Mtn	1.6 ± 3.5	3
JR84-1	36.797	116.462	5/7	213	5.3	173.7	-37.9	17/3	175.9	-39.0	Yucca Mtn	8.1 ± 6.0	3
JR84-5	36.788	116.481	5/8	120	7.0	178.0	-32.0	10/11	185.0	-33.6	Yucca Mtn	17.2 ± 7.2	3
JR84-6	36.785	116.483	6/6	717	2.7	173.7	-36.5	350/14	183.6	-34.4	Yucca Mtn	15.8 ± 3.7	3
JR84-3	36.776	116.461	7/8	578	2.5	169.9	-39.5	27/8	175.8	-44.0	Yucca Mtn	8.0 ± 3.8	3
JR84-7	36.763	116.488	6/7	620	2.7	175.2	-38.8	33/9	181.7	-43.9	Yucca Mtn	13.9 ± 4.0	3
JR87-6	36.753	116.459	7/8	208	4.2	168.1	-35.4	12/17	180.9	-40.6	Yucca Mtn	13.1 ± 5.1	3
8MH-1	36.742	116.441	6/8	637	2.7	161.0	-30.2	40/15	167.2	-42.7	Yucca Mtn	-0.6 ± 3.9	3
JR84-8	36.740	116.456	6/7	200	4.8	170.5	-31.0	27/25	188.4	-43.3	Yucca Mtn	20.6 ± 5.9	3
JR87-8	36.730	116.460	8/9	207	3.9	156.1	-18.3	24/33	183.5	-47.2	Yucca Mtn	15.7 ± 5.3	3
8MH-8	36.724	116.518	8/8	203	3.9	193.2	-33.3	25/12	201.3	-34.9	Yucca Mtn	33.5 ± 4.6	3
8MH-2	36.722	116.441	7/9	322	3.4	161.9	-40.8	7/8	168.4	-43.8	Yucca Mtn	0.6 ± 4.6	3
8MH-11	36.718	116.552	8/10	438	2.7	179.3	-43.7	340/24	196.2	-32.3	Yucca Mtn	28.4 ± 3.6	3
JR87-9	36.718	116.467	7/8	525	2.6	171.7	-30.2	35/12	186.0	-44.0	Yucca Mtn	18.2 ± 3.9	3
8MH-12	36.709	116.563	6/8	682	2.6	184.1	-45.7	5/15	199.0	-44.0	Yucca Mtn	31.2 ± 3.6	3
8MH-9	36.714	116.529	8/8	611	2.2	180.8	-38.5	35/12	190.1	-44.5	Yucca Mtn	22.3 ± 3.6	3
JR87-10	36.705	116.467	7/8	1953	1.4	182.1	-35.0	38/11	189.4	-40.9	Yucca Mtn	21.6 ± 3.0	3
JR87-11	36.693	116.480	8/8	403	2.8	183.2	-44.1	7/13	195.7	-43.5	Yucca Mtn	27.9 ± 4.0	3
8MH-5*	36.692	116.531	6/6	-	8.5	177.1	-47.0	350/15	191.8	-43.2	Yucca Mtn	23.8 ± 9.7	3
JR84-10	36.380	116.491	7/7	337	3.3	184.3	-35.3	21/14	194.6	-35.3	Yucca Mtn	26.8 ± 4.1	3
8BH-5	36.909	116.879	8/8	562	2.3	197.8	-54.2	12/46	239.0	-31.4	Bullfrog Hills	71.2 ±3.4	1
9BH-9	36.951	116.885	7/7	67	7.5	142.5	-14.7	8/61	178.6	-46.6	Bullfrog Hills	10.8 ± 9.1	1
91BH-7	36.185	116.729	8/8	139	4.7	196.8	-25.6	208/8	192.8	-23.5	north of Sleeping Butte	25.0 ± 4.8	1
							Rhyolite of Delirium Canyon						
91YF-44	37.194	116.396	7/8	663	2.4	177.4	-62.7	340/2	181.0	-62.0	southern Pahute Mesa		1
							Yucca Mountain Tuff						
M79-23	36.876	116.460	7/7	442	2.9	184.9	-37.5	76/5	186.3	-42.2	northern Yucca Mtn		4
JR81-3	36.892	116.458	8/8	2612	1.1	179.1	-38.0	22/3	181.3	-39.1	northern Yucca Mtn		4
Average			2						183.8	-40.7			
							Rhyolite of Z						
JR89-21	36.908	116.485	7/7	147	5.0	175.6	-34.2	47/10	180.8	-41.7	northern Yucca Mtn		1
							Pah Canyon Tuff						
M79-17	36.892	116.440	6/6	406	3.3	149.1	-61.9	3/9	165.7	-65.8	north of Yucca Mtn		4
JR81-4	36.892	116.455	7/7	795	2.1	157.7	-61.4	22/3	162.0	-63.4	northern Yucca Mtn		4
Average			2						163.9	-64.6			
							Topopah Spring Tuff						
8YF-43	36.959	116.216	7/7	518	2.7	310.1	70.2	0/0	310.1	70.2	Shoshone Mtn		3
9YF-34	36.993	116.223	8/8	208	3.9	322.0	72.3	205/2	319.4	70.5	Shoshone Mtn		2
JR89-2	36.902	116.488	8/8	355	2.9	299.5	55.1	30/12	299.3	67.1	northern Yucca Mtn		3
Average			3	400	6.2				309.1	69.5			

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TABLE 2 (continued)

Site	Lat°N	Long°W	N/N_o	k	α_{95}	Dg	Ig	St/Dp	Ds	Is	Locality	Rotation	Source
							Tuff of Blacktop Buttes						
9YF-8	37.177	116.360	8/8	756	2.0	354.7	47.6	325/2	356.6	46.6	Blacktop Buttes		1
							Prow Pass Tuff						
JR87-1	36.908	116.494	17/18	216	2.4	348.9	30.9	25/18	0.1	40.1	northern Yucca Mtn		3
							Bullfrog Tuff, Including Stockade Wash Lobe						
JR87-2	36.908	116.498	8/8	389	2.8	358.0	30.9	25/18	11.4	41.3	northern Yucca Mtn	0.4 ± 5.1	3
8YF-27	37.163	116.211	10/10	470	2.2	12.3	48.1	244/4	9.8	44.9	Rainier Mesa	-1.2 ± 4.8	1
8YF-51	37.098	116.210	6/8	200	4.8	13.3	41.5	225/2	11.8	40.4	Big Butte	0.8 ± 6.5	1
Average			3	1047	3.8				11.0	42.2			
8YF-6	36.969	115.927	8/8	959	1.8	13.1	34.9	170/17	0.5	39.9	Halfpint Range	-10.5 ± 4.5	1
8YF-7	36.957	115.946	8/8	328	3.1	14.0	36.8	170/17	0.6	41.9	Halfpint Range	-10.4 ± 5.3	1
91YF-20	36.933	115.874	7/8	323	3.4	359.4	44.7	200/4	355.8	43.2	Halfpint Range	-15.2 ± 5.5	1
91YF-24	36.981	115.866	7/8	457	2.8	358.5	42.9	256/3	357.9	40.0	Halfpint Range	-13.1 ± 5.0	1
92YF-5	36.423	116.239	8/8	266	3.4	14.1	61.4	298/32	20.4	29.9	Little Skull Mtn	9.4 ± 5.2	1
92YF-8	36.420	116.238	8/8	345	3.0	29.1	46.4	225/14	16.6	41.1	Little Skull Mtn	5.6 ± 5.2	1
JR87-14	36.743	116.448	10/10	1136	1.4	4.8	48.0	5/19	24.7	44.7	Yucca Mtn	13.7 ± 4.4	3
8MH-3	36.686	116.535	8/8	218	3.8	14.8	46.0	350/15	26.6	38.3	southern Yucca Mtn	15.6 ± 5.6	3
92MH-1	36.702	116.542	22/23	747	1.1	15.5	50.3	342/24	32.0	34.1	southwest Yucca Mtn	21.0 ± 4.2	1
8BH-4	36.906	116.880	6/7	356	3.6	74.7	73.1	333/44	66.9	29.4	Bullfrog Hills	55.9 ± 5.3	1
9BH-5	36.917	116.928	7/7	408	3.0	283.9	84.4	9/51	98.3	44.6	Bullfrog Hills	87.3 ± 5.3	1
9BH-7	36.953	116.875	7/7	1520	1.6	334.0	35.2	1/60	24.3	37.6	Bullfrog Hills	13.3 ± 4.4	1
9BH-10	36.951	116.932	6/7	95	6.9	181.0	76.8	195/65	270.3	29.4	Bullfrog Hills	-100.7 ± 7.6	1
9BH-1	36.803	116.912	8/8	197	4.0	36.6	42.4	225/42	9.1	25.6	Daylight Pass	-1.9 ± 5.4	1
							Grouse Canyon Tuff						
9YF-11	37.252	116.057	9/9	184	3.8	353.4	14.7	207/2	353.5	12.7	Oak Spring Butte	1.2 ± 5.4	1
8YF-41	37.212	116.128	6/7	366	3.5	353.9	12.2	241/10	353.4	3.0	Whiterock Spring	1.1 ± 5.2	1
8YF-28	37.163	116.198	8/8	459	2.6	350.6	13.7	80/6	350.6	19.7	Rainier Mesa	-1.7 ± 4.9	1
90YF-5	37.196	116.195	6/6	632	2.7	357.4	17.3	229/11	355.9	8.6	Rainier Mesa	3.6 ± 4.9	1
91YF-1	37.183	116.265	9/10	133	4.5	352.1	12.4	40/4	352.8	15.4	Big Burn Valley	0.5 ± 5.8	1
8YF-50	37.096	116.212	7/7	213	4.2	350.4	9.1	220/4	350.1	6.0	Big Butte	-2.2 ± 5.5	1
Average			6	153	5.4				352.3	10.9			
91YF-32	37.241	116.198	7/8	186	4.4	353.8	22.2	326/13	357.7	15.7	Gold Meadows	5.4 ± 5.7	1
91BH-13	37.288	116.970	8/8	105	5.4	355.5	13.0	10/18	13.6	14.6	east of Scottys Junction	21.3 ± 6.3	1
92KV-9	37.371	116.152	7/8	221	4.1	0.1	18.8	13/9	3.3	20.6	Belted Range	11.0 ± 5.6	1
							Comendite of Split Ridge--Upper Lava						
9YF-5	37.166	116.432	8/8	360	2.9	344.4	63.9	13/18	24.3	66.8	Blacktop Buttes		1
							Comendite of Split Ridge--Lower Lava						
9YF-6	37.165	116.342	6/6	266	4.1	352.6	36.9	13/18	6.9	41.1	Blacktop Buttes		1
							Lava of Tram Ridge--Upper Lava						
9YF-4	37.169	116.348	5/7	210	5.3	117.7	-63.3	19/9	121.8	-72.0	Blacktop Buttes		1
							Lava of Tram Ridge--Lower Lava						
9YF-3	37.170	116.350	8/8	80	6.2	123.5	-51.4	19/9	127.2	-60.0	Blacktop Buttes		1
							Quartz-Sanidine Tuff						
9YF-17	37.089	116.194	8/8	37	9.3	13.4	62.7	185/12	350.3	62.0	Tippipah Point		1
							Unnamed Bedded Tuff						
8YF-46	37.093	116.207	7/7	506	2.7	359.9	63.4	220/4	354.4	60.7	Big Butte		1

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TABLE 2 (continued)

Site	Lat°N	Long°W	N/N_o	k	α_{95}	Dg	Ig	St/Dp	Ds	Is	Locality	Rotation	Source
							Comendite of Quartet Dome						
91YF-2	37.216	116.277	8/8	457	2.6	346.4	45.7	40/4	349.1	48.9	Big Burn Valley		1
							Tub Spring Tuff						
9YF-12	37.245	116.057	5/7	418	3.8	349.9	35.9	110/3	348.7	38.5	Oak Spring Butte		1
91YF-33	37.199	116.192	8/8	277	3.3	356.9	41.4	0/0	356.9	41.4	Rainier Mesa		
Average			2						352.8	40.0			
							Unnamed Bedded Tuff						
9YF-1	37.163	116.361	8/8	195	4.0	339.5	61.0	295/15	352.4	50.0	Blacktop Buttes		1
							Unnamed Bedded Tuff						
9YF-33	37.248	116.077	7/9	149	5.0	348.9	57.7	305/4	353.1	54.8	Oak Spring Butte		1
							Tuff of Yucca Flat						
9YF-21	37.015	116.152	6/8	90	7.1	201.9	-41.3	179/11	192.1	-44.7	Mine Mtn		1
							Redrock Valley Tuff						
8YF-1	37.021	116.144	4/8	266	5.6	174.0	-40.5	170/15	161.2	-39.9	Mine Mtn	-0.8 ± 7.6	1
8YF-2	37.022	116.144	7/7	347	3.3	178.7	-36.7	170/15	167.3	-37.5	Mine Mtn	5.3 ± 5.9	1
8YF-3	37.021	116.144	7/7	443	2.9	174.4	-39.5	170/15	162.0	-39.0	Mine Mtn	-0.0 ± 5.4	1
9YF-22	37.019	116.145	8/8	125	5.0	171.1	-43.6	165/15	156.8	-43.3	Mine Mtn	-5.2 ± 7.3	1
8YF-40	37.210	116.128	5/8	184	5.7	168.4	-53.8	241/10	165.3	-44.2	Whiterock Spring	3.3 ± 8.0	1
8YF-45	37.083	116.205	5/8	69	9.3	160.6	-34.2	220/4	159.3	-30.7	Big Butte	-2.7 ± 9.9	1
Average			6	204	4.7				162.0	-39.2			
							Fraction Tuff						
8YF-42	37.208	116.129	7/8	201	4.3	170.1	-42.9	241/10	167.7	-33.4	Whiterock Spring		1
9YF-13	37.243	116.052	8/8	263	3.4	159.8	-41.1	210/2	158.7	-39.6	Oak Spring Butte		1
Average			2						163.2	-36.5			
							Basalt of Whiterock Spring						
9YF-24	37.200	116.131	8/8	489	2.5	3.3	59.8	233/12	350.4	50.0	Whiterock Spring		1
							Tuff of Whiterock Spring						
9YF-15	37.239	116.043	8/8	177	4.2	149.2	-58.8	47/3	150.4	-61.7	Oak Spring Butte		1
9YF-23	37.198	116.130	7/9	841	2.1	144.3	-55.6	198/20	133.2	-38.3	Whiterock Spring		1
Average			2						141.8	-50.0			

Site is site name; Lat and Long are site latitude and longitude; N/N_o is number of samples accepted/number of samples demagnetized; k is concentration parameter [*Fisher*, 1953]; α_{95} is 95% confidence cone about mean; Dg and Ig are declination and inclination of site mean direction in geographic coordinates; St/Dp are strike and dip of site (dip direction 90° clockwise from strike); Ds and Is are declination and inclination of site mean direction in stratigraphic coordinates; Locality is general geographic area of site; Rotation is estimate of vertical-axis rotation of site relative to average reference direction for individual ash flow tuffs with 95% confidence limits [*Demarest*, 1983]; Sources are 1, this study; 2, *Hudson* [1992]; 3, *Rosenbaum et al.* [1991]; and 4, *Rosenbaum and Synder* [1985]. * Site mean derived from remagnetization circle analysis; α_{95} estimated as average of maximum and minimum axes of confidence ellipse. Mtn is mountain. Average is unit mean of multiple sites from the central southwestern Nevada volcanic field.

trapped the same geomagnetic field direction. Results from the Bullfrog and Stockade Wash units are considered together in subsequent analyses.

Comparison to the Craton

As there is no evidence for rotation among localities, remanence directions from units having local distribution in the central SWNVF can be combined with data from widespread ash flow tuffs to derive a tightly constrained time-averaged composite mean direction, which is D = 351.4°, I = 52.7°, α_{95} = 4.5°, n=32 (Figure 4a and Tables 2 and 3). Low-inclination remanent magnetizations carried by Trail Ridge Tuff and the Grouse Canyon Tuff (Figure 3) probably represent transitional field directions and were excluded from calculation of this composite mean.

The accuracy of the time-averaged direction for the central SWNVF, and of a rotation estimate calculated from this direction (below), depends on the degree to which the composite mean has averaged paleosecular variation. Paleosecular variation may be difficult to sample and thus average for volcanic sequences that erupted in pulses that were rapid with respect to the paleosecular variation rate [e.g., *Mankinen et al.*, 1987;

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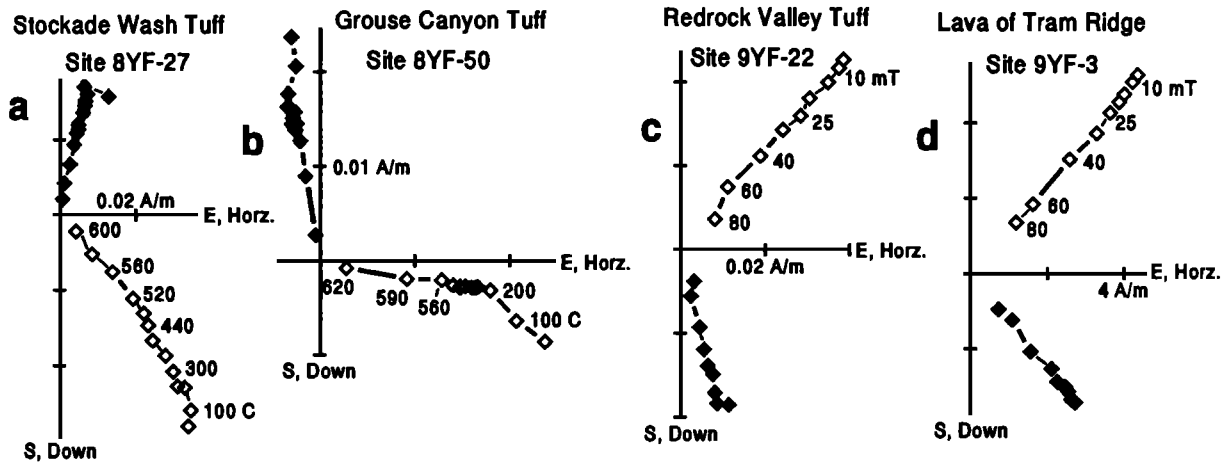


Figure 2. Representative orthogonal diagrams from different rock units of the southwestern Nevada volcanic field depicting their characteristically uncomplicated demagnetization behavior. Solid and open symbols are projections on horizontal and vertical planes, respectively. Examples from several additional, widespread ash flow tuffs are given by *Rosenbaum et al.* [1991] and *Hudson* [1992].

Calderone et al., 1990; *Hudson and Geissman*, 1991]. Clustering of $^{40}Ar/^{39}Ar$ ages for different units from the major stratigraphic groups of the field (Table 1) indicates that SWNVF rocks erupted discontinuously during the life of the field [*Sawyer et al.*, 1990, in press]. Nonetheless, units from the major SWNVF stratigraphic groups typically carry directions that are distinct from stratigraphically adjacent units, indicating that they did record independent paleofield directions.

The dual polarities of magnetizations recorded by the composite sequence also suggests that the composite sequence adequately averaged paleosecular variation. Mean directions for the normal and reverse polarity units (Figure 4b and Table 3) yield a positive reversal test [*McFadden and Lowes*, 1981]. Units of the composite sequence span at least nine magnetic polarity reversals. Preliminary correlation (Figure 5) of the SWNVF reversal sequence to the magnetic polarity timescale of *Harland et al.* [1990] suggests that the composite volcanic sequence sampled at least 14 polarity chrons or subchrons. Detailed correlation of the SWNVF magnetostratigraphy to published magnetic polarity timescales is a subject of ongoing study.

The angular dispersion (S value of *Doell* [1970]) of virtual geomagnetic poles (VGPs) for the central SWNVF composite sequence is 18.1° with lower and upper 95% confidence limits [*Cox*, 1969] of 15.5° and 21.8°, respectively. This angular dis-

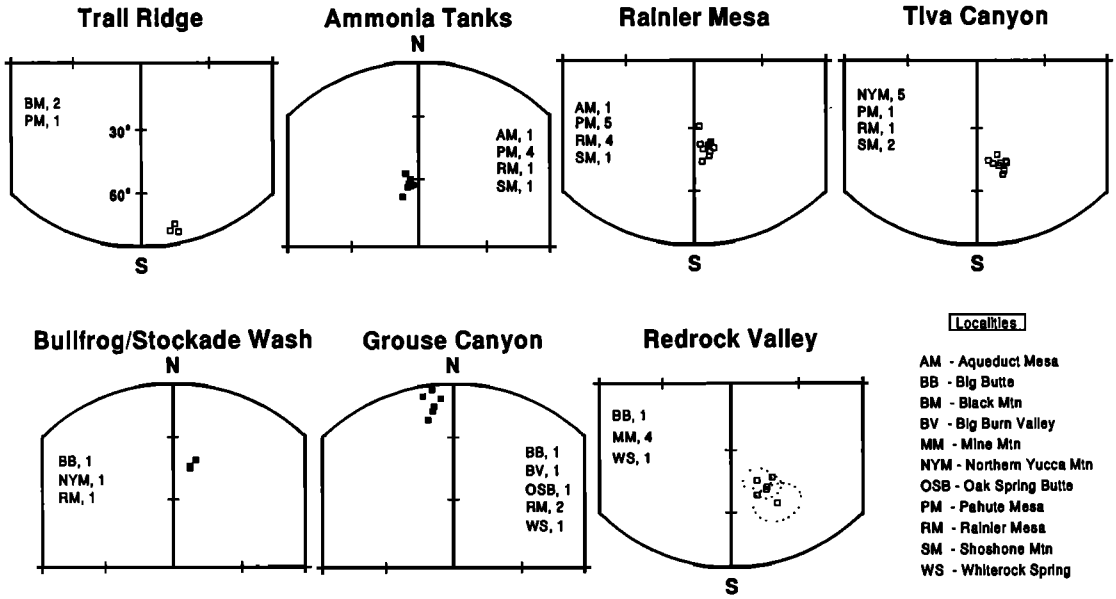


Figure 3. Equal-area projections depicting tilt-corrected mean remanence directions for individual sites from seven widespread ash flow cooling units. The general locality (e.g., PM, Pahute Mesa) and the number of sites sampled at that locality are listed for each tuff. Localities are shown in Figures 1 and 6. Solid and open symbols are projections on the lower and upper hemispheres, respectively. The 95% confidence cones greater than 5° are shown by dotted lines.

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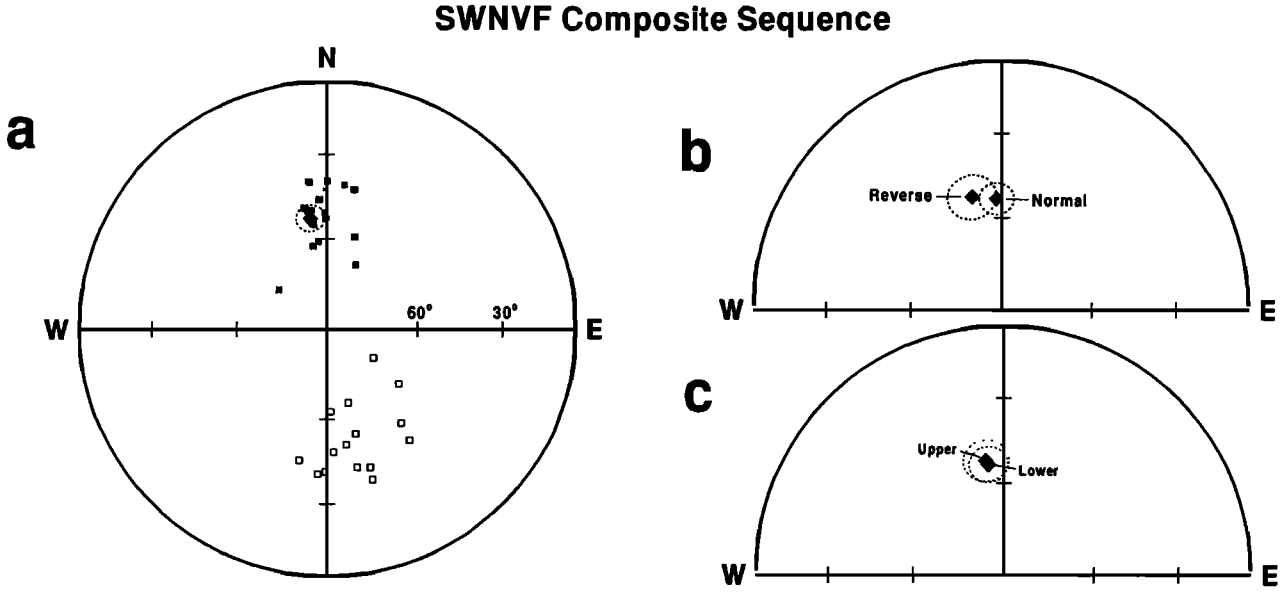


Figure 4. Equal-area projections showing (a) tilt-corrected unit directions (squares, Table 2) and time-averaged mean direction (diamond) for the composite sequence from the central southwestern Nevada volcanic field, (b) mean direction of normal polarity units and inverted mean direction of reverse polarity units, and (c) mean directions of lower and upper halves of the composite sequence. Dotted lines are 95% confidence cones around mean directions. Solid and open symbols are projections on the lower and upper hemispheres, respectively.

persion is similar to a 19.3° value predicted for this latitude based on a model of paleosecular variation of 5- to 45-Ma rocks by *McFadden and McElhinny* [1984]. Considered together, this level of VGP angular dispersion, the multiple polarity intervals sampled, and the positive reversal test indicate that the central SWNVF composite mean has adequately averaged paleosecular variation.

The direction of the time-averaged mean for the central SWNVF is similar to that predicted for the area from published middle Miocene paleopoles for North America, but detailed comparisons vary depending on the paleopole used. Paleopoles and expected directions considered are (1) the 10-Ma paleopole of the 30-m.y.-window analysis of *Irving and Irving* [1982], yielding an expected direction of D = 358.5°, I = 55.1°, (2) a paleopole (87.5° N, 194.6° E, A_{95} = 3.5°) calculated from 7- to 15-Ma rock units listed in Table 1 (entries 10-17) of *Irving and Irving* [1982], yielding D = 357.6°, I = 58.2°, and (3) the 10-Ma paleopole calculated from the 20-m.y.-window analysis of the master apparent polar wander path of *Besse and Courtillot* [1991], yielding D = 352.5°, I = 56.4°. Table 3 gives rotation and flattening parameters [*Beck*, 1980; *Demarest*, 1983] based on comparisons of the central SWNVF mean to these expected directions. Of these, we prefer the parameters (R = -7.1° ± 6.6° and F = 2.4° ± 3.6°) based on the 10-Ma paleopole of *Irving and Irving* [1982] because this paleopole more closely falls between the present rotation axis of the Earth and Oligocene to early Miocene paleopoles of *Harrison and Lindh* [1982], *Hagstrum et al.* [1987], or *Diehl et al.* [1988] than does the 10-Ma palepole of *Besse and Courtillot* [1991]. This comparison suggest that the central SWNVF rotated counterclockwise a small amount

Table 3. Mean Directions and Rotation Estimates for the Central Southwestern Nevada Volcanic Field Composite Sequence

	N	D	I	k	α_{95}	Res	Lat	Long	δp	δm	Paleopole 1 R	Paleopole 1 F	Paleopole 2 R	Paleopole 2 F	Paleopole 3 R	Paleopole 3 F
All units	32	351.4	52.7	32	4.5	31.0426	82.0	126.9	4.3	6.2	-7.1 ± 6.6	2.4 ± 3.6	-6.2 ± 7.0	5.5 ± 3.6	-1.1 ± 6.4	3.7 ± 3.6
Upper	15	352.0	53.4	40	6.1	14.6531	82.7	129.3	5.9	8.5	-6.5 ± 8.6	1.7 ± 4.8	-5.6 ± 8.9	4.8 ± 4.8	-0.5 ± 3.0	3.0 ± 4.8
Lower	17	351.0	52.0	26	7.1	16.3922	81.3	124.3	6.6	9.7	-7.5 ± 9.6	3.1 ± 5.6	-6.6 ± 9.8	6.2 ± 5.6	-1.5 ± 9.4	4.4 ± 5.6
Normal	17	357.3	53.4	44	5.4	16.6401	86.1	98.3	5.2	7.5	-1.2 ± 7.8	1.7 ± 4.3	-0.3 ± 8.0	4.8 ± 4.3	4.8 ± 7.5	3.0 ± 4.3
Reverse	15	345.0	51.5	27	7.6	14.4723	76.7	135.7	7.0	10.3	-13.5 ± 10.1	3.6 ± 6.0	-12.6 ± 10.3	6.7 ± 6.0	-7.5 ± 9.9	4.9 ± 6.0

N is number of units; D and I are declination and inclination of mean direction, respectively; k is concentration parameter [*Fisher*, 1953]; α_{95} is cone of 95% confidence about mean direction; Res is resultant vector; Lat and Long are paleopole latitude and longitude, respectively; δp and δm are axes of 95% confidence ellipse about paleopole; R and F are rotation and flattening parameters, respectively, based on comparison of mean direction with to an expected direction derived from 1, 10 Ma paleopole from 30-m.y. window analysis of *Irving and Irving* [1982]; 2, paleopole calculated from entries 10-17 of Table 1 of *Irving and Irving* [1982]; and 3, 10 Ma paleopole from *Besse and Courtillot* [1991]. 95% confidence limits are after *Demarest* [1983].

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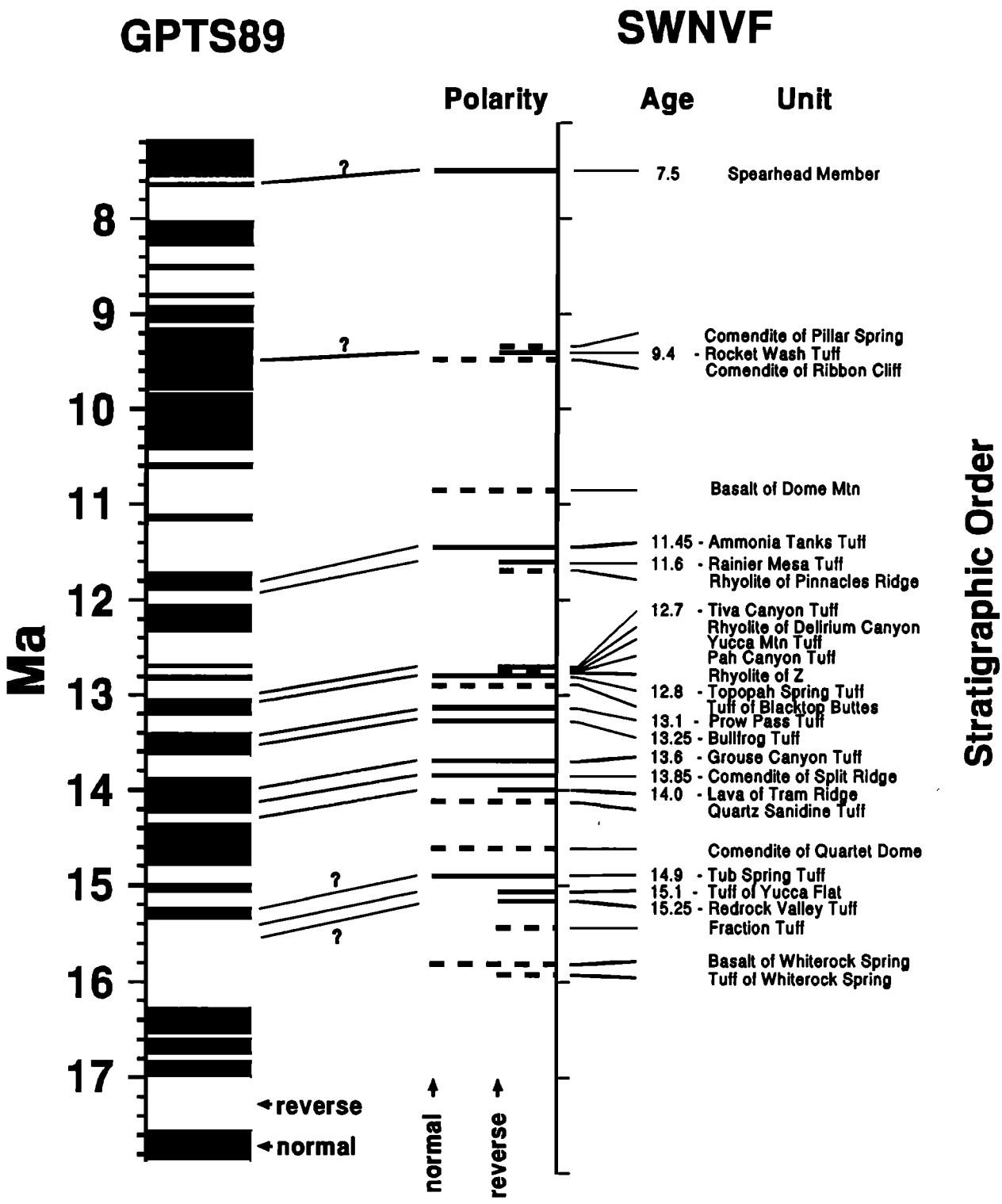


Figure 5. Comparison of the composite magnetic reversal sequence for the southwestern Nevada volcanic field to the magnetic polarity timescale from *Harland et al.* [1990]. For SWNVF polarity column, solid lines represent units with ages controlled by $^{40}Ar/^{39}Ar$ ages [*Sawyer et al.* 1990, in press] whereas dashed lines represent units whose ages are estimated from the stratigraphic stacking order and petrologic affinities. Lines between the SWNVF reversal sequence and the polarity timescale are our preferred correlations but are preliminary. Queries mark correlations of greatest uncertainty. The age shift inferred between the two columns is controlled mainly by polarity changes between Ammonia Tanks and Rainier Mesa tuffs, Tiva Canyon and Topopah Spring tuffs, and comendite of Split Ridge and lava of Tram Ridge. This comparison suggests that remanent magnetizations carried by units of the SWNVF composite sequence recorded at least 14 different polarity chrons or subchrons.

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with respect to the North American craton. Little rotation (-1.1° ± 6.4°) is implied from the comparison with the *Besse and Courtillot* [1991] paleopole.

Because tectonism was concurrent with middle Miocene development of the SWNVF both locally and regionally in the southern Great Basin, it is also important to test for any rotation through time. The mean for the half of the composite sequence younger than the 13.25-Ma Crater Flat Group is D = 352.0°, I = 53.4°, α_{95} = 6.1°, n = 15, whereas that for the Crater Flat and older half is D = 351.0°, I = 52.0°, α_{95} = 7.1°, n = 17. These mean directions are statistically indistinguishable from each other and from the mean of the full composite sequence [*McFadden and Lowes*, 1981], indicating that no significant vertical-axis rotation affected the central SWNVF during development of the field.

Rotation Constraints for Outlying Areas

The sense and amount of vertical-axis rotation at individual sites of widespread ash flow tuffs, particularly from outlying areas of the field, can be assessed by comparing the site means to unit means derived from the central SWNVF. Figure 6 illustrates vertical-axis rotation estimates calculated in this manner for numerous sites across the volcanic field. These rotation estimates typically have uncertainty limits less than 10° (Table 2), but these uncertainties do not incorporate possible tilt-correction errors for the individual sites. Because of these potential errors, we place confidence in rotations of less than 15° only when they have been replicated at several sites from an area. Our confidence in rotations also increases when similar rotations are obtained from different units in a local area. Conclusions gained from the compilation of Figure 6 are summarized below.

Results for two sites from Pahute Mesa southwest of Kawich Valley are exceptions to a general conclusion that individual sites from the central SWNVF were little affected by vertical-axis rotation. These two sites from Ammonia Tanks and Rainier Mesa tuffs have declinations that suggest clockwise rotations of 20° and 40°, respectively. These sites lie west of the Kawich Valley fault, a range-front normal fault whose offset decreases southward. Clockwise rotation west of the southern Kawich Valley fault might arise from a transfer of extensional strain to faults farther to the southwest in Pahute Mesa (not shown, but see *Orkild et al.* [1969]). Further sampling is required to verify that clockwise rotation is spatially associated with the Kawich Valley fault.

In the eastern part of the volcanic field, *Hudson* [1992] concluded that arcuate trends of faults that offset tuffs in the Massachusetts Mountain area were not due to a 90° oroclinal bend, as had been previously proposed. Negligible rotation was found for sites from the southwestern Halfpint Range. Additional data from this study from Ammonia Tanks (one site), Rainier Mesa (one site), and Bullfrog (four sites) tuffs, however, suggest that the southeastern Halfpint Range rotated counterclockwise about 15°. The pattern of rotations suggests that normal fault blocks extended in a fanlike fashion within the southern Halfpint Range.

Clockwise rotation of approximately 25° to 30° of the southern part of Yucca Mountain relative to the northern part was previously described by *Rosenbaum et al.* [1991], mostly from data from the Tiva Canyon Tuff. This study illustrates that northern Yucca Mountain has not been rotated relative to a large area of the central SWNVF and that this area has undergone minimal rotation with respect to the North American craton. Our findings support the conclusion of *Rosenbaum et al.* [1991] that the more extended southern part of Yucca Mountain rotated clockwise. Rotations for the Tiva Canyon Tuff in this paper are 3.4° less than those presented by *Rosenbaum et al.* [1991] because they are based on a comparison to an average Tiva Canyon reference declination rather than to the declination from a single site mean from northern Yucca Mountain.

An angular unconformity between the Timber Mountain Group and older volcanic rocks at Yucca Mountain may mark the onset of clockwise rotation. Whereas an average of nine site mean directions from the Tiva Canyon Tuff from the southern part of Yucca Mountain indicates vertical-axis rotation of 26.1° ± 4.8°, an average of seven site mean directions from the Ammonia Tanks Tuff from this area suggests vertical-axis rotation of only 6.2° ± 12.7° (Figure 7). Individual declinations for sites from the Ammonia Tanks Tuff, however, show considerable scatter (Figures 6 and 7). There was surface relief related to pre-11.6-Ma faulting in this area at the time of deposition of Timber Mountain units [*Carr*, 1984; *Scott*, 1990]. Much dispersion of the Ammonia Tanks declinations may reflect influence of primary foliation dips on tilt corrections for the sites. Nonetheless, individual rotation estimates for four sites cluster near the overall Ammonia Tanks rotation mean. These data suggest that clockwise rotation of southern Yucca Mountain began before 11.45 Ma and proceeded at a minimum rate of about 16°/m.y.

Rotation for one site from 11.6-Ma Rainier Mesa Tuff from southern Yucca Mountain (26.7° ± 5.5°) is similar to the average Tiva Canyon rotation (Figure 7). If rotation for this Rainier Mesa site is valid, the rate of rotation between deposition of the Rainier Mesa and Ammonia Tanks tuffs was extreme (>130°/m.y.). We do not currently have confidence in this interpretation, however, because the Rainier Mesa site mean might be affected by dispersion similar to the Ammonia Tanks site means.

Elsewhere in the southern SWNVF, only small rotations (≈ 10°) are indicated for several sites from Ammonia Tanks, Rainier Mesa, and Bullfrog tuffs near the Rock Valley fault. Because of anomalous northeast to east-northeast strikes of Tertiary faults and pre-Tertiary folds and thrusts, numerous workers [*Burchfiel*, 1965; *Ekren*, 1968; *Barnes et al.*, 1982; *Wernicke et al.*, 1982; *Snow*, 1992; *Duebendorfer and Black*, 1992] have proposed that this area was affected by large clockwise oroclinal rotation (as great as 90°) related to dextral movement on the Las Vegas Valley shear zone during Tertiary deformation. Our results suggest that if large rotation affected this region, it mostly predated deposition of these tuffs at 11.45 Ma, 11.6 Ma, and 13.25 Ma. Miocene volcanic rocks in the Rock Valley fault area are clearly deformed, however, being broken by numerous normal and strike-slip faults and locally having dips as great as 60° (e.g., at Pink Holes Hill [*Hinrichs*, 1968]). Thus far no paleomagnetic studies have been completed to test for rotation of Paleozoic rocks in the Specter and Spotted ranges, but *Gillett and Van Alstine* [1982] and *Nelson and Jones* [1987] documented large clockwise rotations adjacent to the Las Vegas Valley shear zone farther to the southeast.

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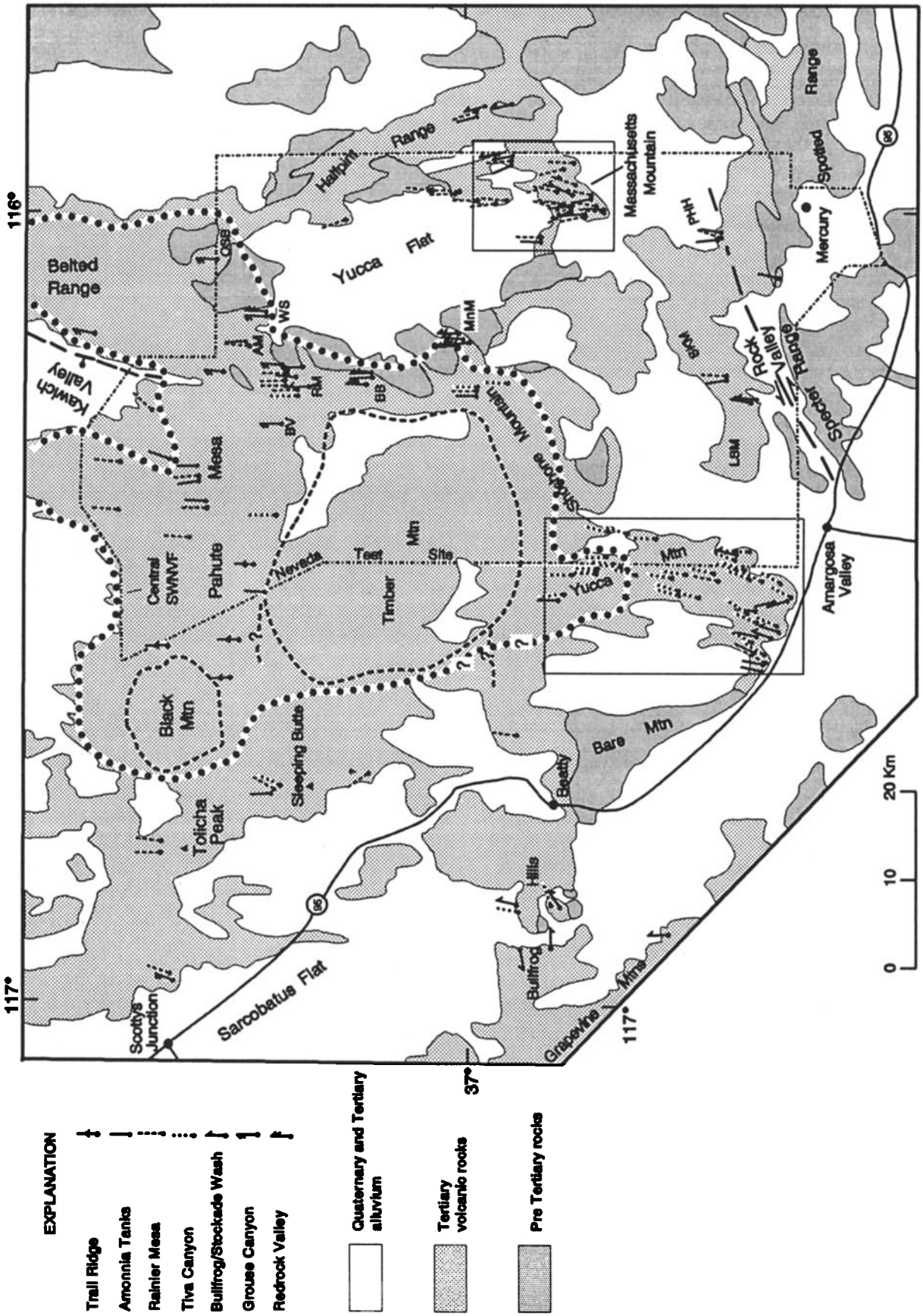


Figure 6. (continued)

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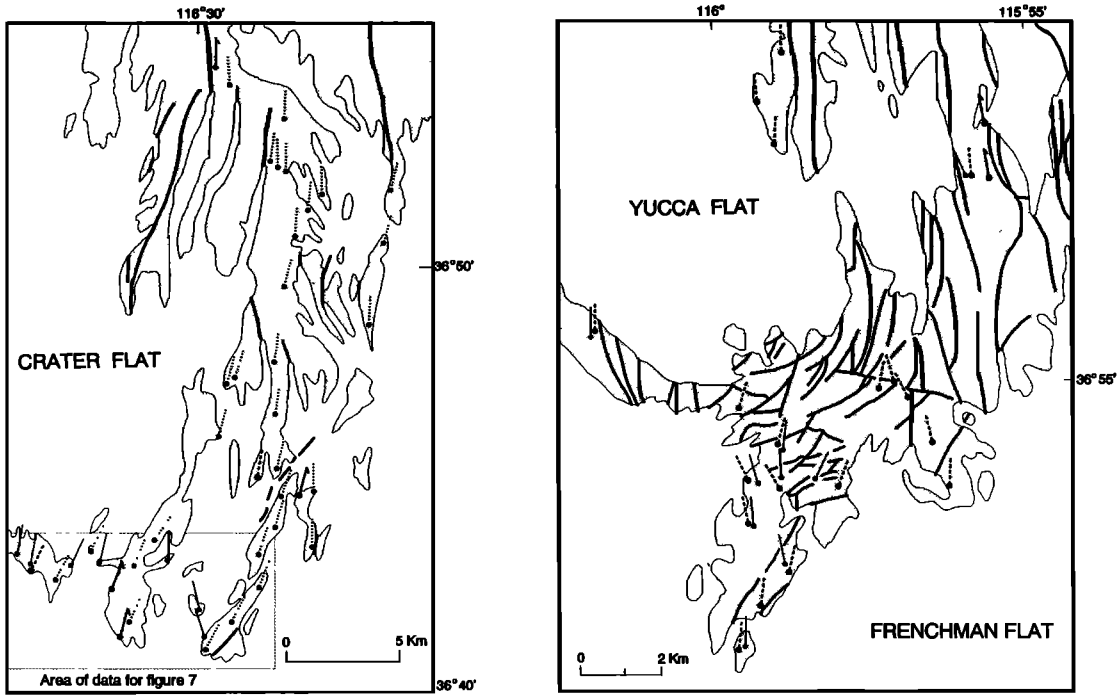


Figure 6. Simplified geologic maps of southwestern Nevada showing estimates of relative vertical-axis rotation for numerous sites (dots) from seven ash flow cooling units from the southwestern Nevada volcanic field (SWNVF). Sense and magnitude of rotations are shown by deflection of the lines from geographic north. Rotations are calculated from comparison of tilt-corrected site mean declinations to appropriate average reference declinations for the seven units (Table 2). Dotted line marks approximate boundary of the central SWNVF area affected by relatively low tectonic strain. Abbreviations are AM, Aqueduct Mesa; BB, Big Butte; BV, Big Burn Valley; LSM, Little Skull Mountain; MnM, Mine Mountain; OSB, Oak Spring Butte; PHH, Pink Holes Hill; RM, Rainier Mesa; SKM, Skull Mountain; and WS, Whiterock Spring

Large but variable rotations are indicated for several sites from the western Bullfrog Hills (Figure 6) near the western margin of the SWNVF. These rocks have been strongly extended above a detachment fault [*Maldonado*, 1990a]. Although the rotations might reflect only small-scale heterogeneity of deformation in the hanging wall of the detachment fault [*Wells and Hillhouse*, 1989], the fact that four of five fault blocks sampled have clockwise rotation implies that deformation in the Bullfrog Hill may have included a component of regional dextral shear strain.

A single site in 13.25-Ma Bullfrog Tuff near Daylight Pass, Grapevine Mountains, indicates negligible rotation. *Snow and Wernicke* [1989], however, suggested that upper plate Paleozoic rocks from a broad region just west of this site may have been rotated as much as 90° counterclockwise on the basis of an anomalous west-northwest trend of a fold.

Paleomagnetic results from the sparsely sampled northwestern SWNVF suggest that moderate clockwise rotation affected this region at least locally. Two sites from east of Scottys Junction and two of three sites from north of Sleeping Butte suggest 20-27° clockwise rotation. Two sites from north of Tolicha Peak, however, indicate negligible rotation.

Discussion

Paleomagnetic data indicate that the central SWNVF experienced at most only small counterclockwise vertical-axis rotation (-7.1° ± 6.6°) relative to the North American craton during or since middle Miocene time. This finding implies that variations in trends of structural features within this area cannot be attributed to middle Miocene or younger vertical-axis rotation. The rotational estimates for the central SWNVF and outlying areas of the SWNVF also place constraints on regional tectonics.

Walker Lane Influence

The degree to which northwest striking dextral faults related to the Walker Lane belt may have deformed and influenced development of the SWNVF is debatable. *Carr* [1984, 1990) proposed an active role for dextral faulting during development of the SWNVF. He suggested that dextral faults may be present far enough northeast to extend beneath Pahute Mesa and across southern Yucca Flat (Figure 8a) and that such faults had sufficient middle Miocene displacement to localize calderas in a pull-apart structure at right-stepping segments of the Walker Lane boundary. Throughgoing dextral faults are lacking at the

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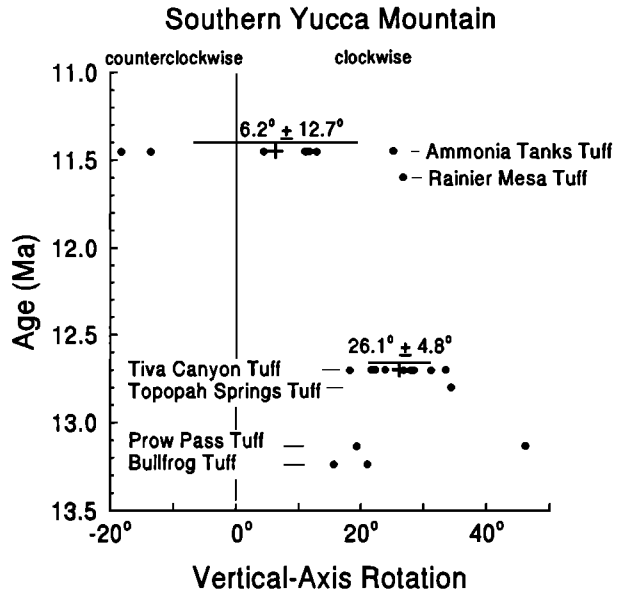


Figure 7. Estimates of vertical-axis rotation versus age of ash flow cooling units from the southern part of Yucca Mountain (see Figure 6 for area). Dots show rotation estimates for individual sites, whereas pluses and horizontal lines show estimates and their uncertainty limits for the averages of individual sites for Tiva Canyon and Ammonia Tanks tuffs. Data for this figure are derived from *Rosenbaum et al.* [1991] and Tables 1 and 2 of this study.

surface in the central SWNVF, and the presence of a buried dextral fault in southern Yucca Flat has been discounted [*Hudson*, 1992]. Nonetheless, Carr's model is difficult to disprove beneath Pahute Mesa because such faults might be largely buried beneath younger Timber Mountain and Thirsty Canyon groups.

Although not predicted by *Carr* [1984, 1990], clockwise vertical-axis rotation might have occurred in rocks of the central SWNVF if they were affected by strong dextral faulting. If deformed concurrently with volcanism, lower and middle crust beneath the central SWNVF would probably have behaved ductilely due to its high temperature. Broadly distributed dextral shear strain at depth would likely cause the overlying brittle upper crust to respond by clockwise vertical-axis rotation of crustal blocks [*McKenzie and Jackson*, 1983; *Sonder et al.*, 1986; *Lamb*, 1987]. Moreover, physical models [*Hempton and Neher*, 1986; *Cobbold et al.*, 1989] and limited natural examples [*Doughty and Sheriff*, 1992] suggest that vertical-axis rotation may be enhanced at dilational jogs between en echelon strike-slip shear zones. Paleomagnetic data from this study, however, indicate that vertical-axis rotation of the central SWNVF with respect to the stable craton was small and, if real, counterclockwise in sense. Moreover, there is no indication of relative rotation between sites that sampled individual ash flow tuffs both outside and within Carr's suggested zone of Walker Lane influence (e.g., compare Grouse Canyon Tuff site means from Oak Spring Butte, White Rock Spring, Rainier Mesa, and Big Burn Valley localities). Lacking clockwise vertical-axis rotation, throughgoing northwest striking surface faults, or evidence for dextral offset on any possible subsurface structures beneath Pahute Mesa [*Warren et al.*, 1985], we find little to support a model of active middle Miocene dextral slip and pull-apart basin development in the central SWNVF. This conclusion does not preclude a scenario in which northwest striking structures beneath Pahute Mesa or elsewhere in the central SWNVF could have originated before but may have experienced minor reactivation during middle Miocene time, as may be true elsewhere in the Walker Lane [*Stewart*, 1988].

The lack of rotation of 11.45- to 13.25-Ma tuffs at the southeast margin of the SWNVF in the Rock Valley area contradicts previous models in which these middle Miocene rocks are involved in clockwise oroclinal bending at the northwest end of the Las Vegas Valley shear zone [*Ekren*, 1968; *Barnes et al.*, 1982; *Wernicke et al.*, 1982, *Snow*, 1992; *Duebendorfer and Black*, 1992]. If clockwise vertical-axis rotation is responsible for the anomalous structural trends in this area, the rotation must be older than some workers have suggested. Considering only age constraints from the Rock Valley area, we cannot preclude the suggestion by *Royse* [1983] that bending along parts of the Las Vegas Valley shear zone might, at least partly, be related to Mesozoic thrusting. We prefer an alternative, however, that rotation was Cenozoic but predates 13.25 Ma for the following reasons. The contrast in amount of extension north and south of the Las Vegas Valley shear zone requires some Cenozoic movement on the shear zone [*Davis and Burchfiel*, 1973; *Guth*, 1981; *Duebendorfer and Black*, 1992]. At the southeast end of the Las Vegas Valley shear zone, *Sonder et al.* [1989] and *Jones et al.* [1991] reported paleomagnetic data indicating 15- to 16-Ma rocks were strongly rotated clockwise. In the Black Hills Basin, which formed during Miocene extension north of the central Las Vegas Valley shear zone [*Guth et al.*, 1988], an angular unconformity near the top of the basin fill separates 30°-dipping rocks from overlying 10°-dipping tephra that is petrographically correlated with the 13.25-Ma Bullfrog Tuff (D. L. Schmitt and R. G. Warren, personal communication, 1991). This relation permits most Miocene extension adjacent to this part of the Las Vegas Valley shear zone to predate 13.25 Ma. Thus clockwise rotation in the Rock Valley area could be Miocene but older than 13.25 Ma. If clockwise rotation was synchronous along the Las Vegas Valley shear zone, these age constraints suggest that deformation associated with the shear zone was short lived but intense.

Cenozoic dextral shear probably did affect the southern and western margins of the SWNVF (Figure 6). This is well documented for southern Yucca Mountain where numerous sites show evidence of about 25° clockwise vertical-axis rotation of 12.7-Ma to 13.25-Ma tuffs [*Rosenbaum et al.*, 1991]. Clockwise rotation at Yucca Mountain began between 12.7 Ma and 11.45 Ma (Figure 7). Clockwise rotation of sites in the Bullfrog Hills and areas farther to the northwest also probably reflect dextral shear which may be linked with deformation at Yucca Mountain. The lack of rotation from the Rock Valley area, however, indicates that dextral shear did not occur synchronously within a single zone that transected the entire southern margin of the volcanic field and connected with the Las Vegas Valley shear to the southeast.

Our study suggests that Cenozoic dextral shear zones that affected the region of the SWNVF were discontinuous in length and diachronous in age. Clockwise rotations at the southern and western margins of the field occurred at the interface between

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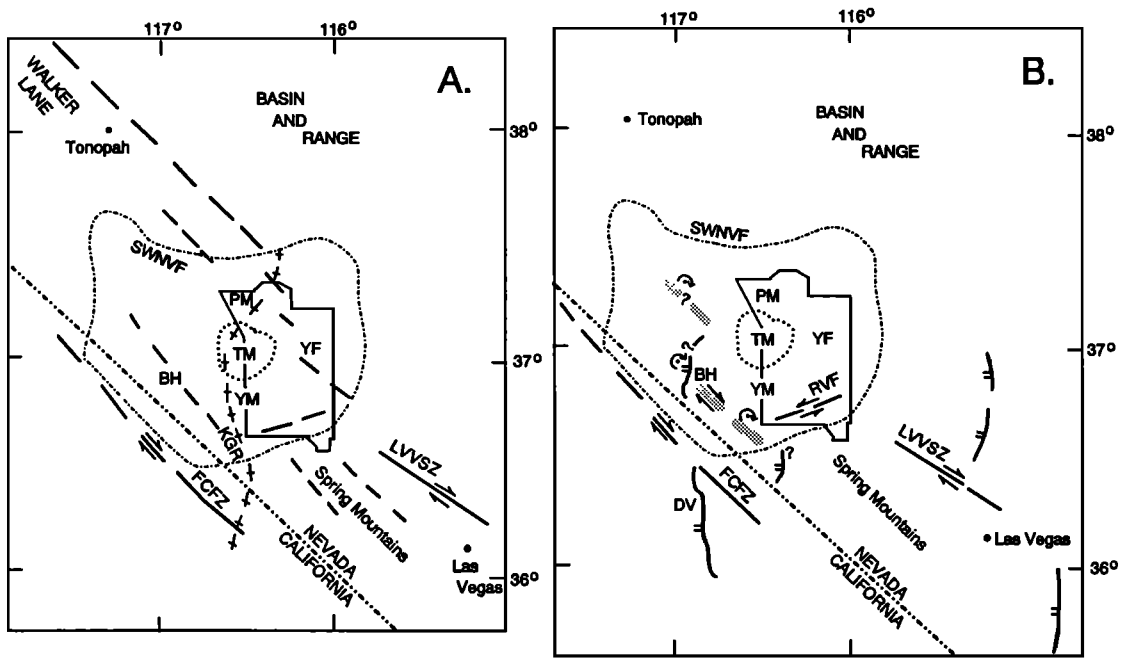


Figure 8. Schematic models for the possible tectonic influence of dextral shearing on southwestern Nevada volcanic field (SWNVF) rocks. (a) Model after *Carr* [1984, 1990] showing broad Walker Lane zone containing numerous northwest striking faults having known or inferred dextral offset. Faults at the northeast boundary of the Walker Lane zone in southern Nevada are inferred to step right, forming the Kawich-Greenwater rift (KGR) concurrently with dextral faulting. (b) Model showing dextral shearing restricted to southwestern regions of the SWNVF on the basis of clockwise vertical-axis rotation occurrence. Faults with double tick marks schematically show areas of strong extension. Dextral strain may be localized at interfaces between the relatively unextended central SWNVF region and more extended regions to the southwest. Dextral shear may be diachronous due to temporal shifts in extension centers. Abbreviations are BH, Bullfrog Hills; DV, Death Valley; FCFZ, Furnace Creek fault zone; KGR, Kawich-Greenwater rift; LVVSZ, Las Vegas Valley shear zone; PM, Pahute Mesa; RVF, Rock Valley fault; TM, Timber Mountain; YF, Yucca Flat; and YM, Yucca Mountain.

the less extended central SWNVF and more extended marginal regions [*Maldonado*, 1990a]. We envision the zones of dextral shear to be in part extensional tranfer zones [*Gibbs*, 1984] that formed to accommodate local, diachronous centers of crustal extension (Figure 8b). This model implies that dextral shear of the southern and western margins of the SWNVF was caused by post-12.7-Ma extension south of Yucca Mountain and west of the Spring Mountains [*Brocher et al.*, 1993], whereas any Cenozoic rotation in the Rock Valley area would have been generated by pre-13.25-Ma extension in areas farther to the east. Rather than occurring on a discrete transfer fault, dextral shear was probably accommodated by a system of obliquely extending fault blocks undergoing a component of vertical-axis rotation across a zone of several kilometers width. Although we think these dextral shear zones acted to transfer localized extension, they also may have accommodated a component of Cenozoic horizontal shortening roughly normal to extension [*Anderson and Barnhard*, 1993].

Province-Scale Counterclockwise Rotation?

Our analysis suggests that the central SWNVF most likely rotated counterclockwise a small amount (-7.1° ± 6.6°) with respect to cratonic North America, although this conclusion depends on the choice of reference paleopole (Table 3). If this rotation is real, the central SWNVF probably rotated as a semirigid block inasmuch as it lacks pervasive strike-slip faults that could internally accommodate rotation of smaller blocks in a "domino" mechanism [e.g., *Ron et al.*, 1984]. Palinspastic reconstructions [*Wernicke et al.*, 1988] suggest that the central SWNVF block was translated several tens of kilometers away from the Colorado Plateau during predominantly middle Miocene and younger extension in the southern Great Basin. It is plausible that the central SWNVF block could have rotated counterclockwise a small amount during such westward translation.

Paleomagnetic studies suggest counterclockwise rotation also may have affected middle Miocene rocks farther to the north (Figure 9) in the Dixie Valley region (-6° ± 10°) [*Hudson and Geissman*, 1991] and the Nevada Rift region (-19° ± 7°) [*Li et al.*, 1990]. If related, counterclockwise rotation at these three widely separated localities may be evidence of a province-wide phenomenon (although *Li et al.* [1990] called on local causes in their rotation model). Contemporary counterclockwise rotation of the Sierra Nevada block detected by *Argus and Gordon* [1991] might be the current expression of this phenomenon. Such widespread rotation could reflect a province-wide strain gradient [*Hamilton and Myers*, 1966] in which northern parts of the province experienced greater cumulative extension since middle Miocene as they translated away from the Colorado Plateau. Such a widespread but small counterclockwise rotation

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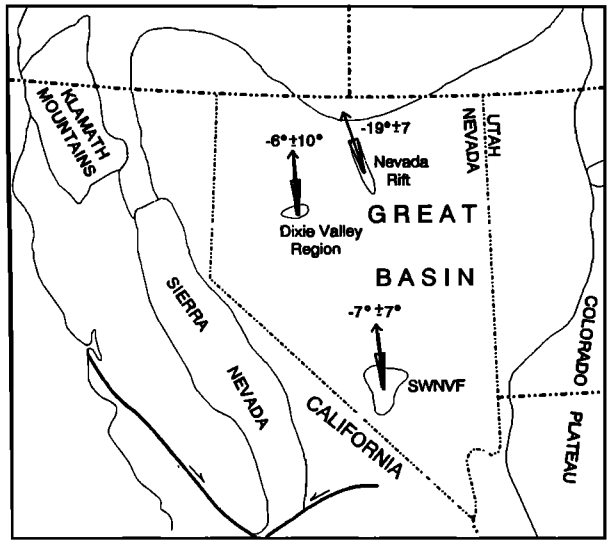


Figure 9. Paleomagnetic indications of small- to moderate magnitude counterclockwise vertical-axis rotation of middle Miocene rocks from the southwestern Nevada volcanic filed (this study), the Dixie Valley region [*Hudson and Geissman*, 1991], and the Nevada rift [*Li et al.*, 1990] regions of the Great Basin.

would probably be superimposed on contemporaneous or older vertical-axis rotations that have more local distributions and causes.

Conclusions

Paleomagnetic data from individual widespread ash flow cooling units ranging in age from about 15 Ma to 9.5 Ma document that localities within a broad area of the central SWNVF underwent no vertical-axis rotation relative to one another. Thus compilation of remanence directions from numerous volcanic deposits from the central SWNVF that trapped independent paleofield directions allows a well constrained time-averaged mean paleomagnetic direction to be calculated. This mean direction (D = 351.4°, I = 52.7°, α_{95} = 4.5°) is similar to a middle Miocene expected direction predicted for the area, indicating that the central SWNVF underwent minimal counterclockwise vertical-axis rotation (R = -7.1° ± 6.6°) with respect to the North American craton.

Some workers have suggested that dextral faults of the Walker Lane zone project beneath the central SWNVF and that they were active during and may have influenced development of the volcanic field. The paleomagnetic data, however, reveal no dextral strain manifest as regional scale clockwise vertical-axis rotation. This result, coupled with the lack of any through-going dextral faults with perceptible offset in the Tertiary volcanic rocks, does not support a hypothesis of active dextral shear beneath the central SWNVF during middle Miocene time.

Referenced to average unit mean directions from the central SWNVF, numerous sites from individual ash flow tuffs along southern and western margins of the field show evidence for moderate clockwise rotation. We suggest that this clockwise rotation accommodated dextral shear strain developed at the interface between the little extended central SWNVF block and more strongly extended areas to the south and southwest. Negligible rotation, however, was found for 11.45- to 13.25-Ma tuffs at the southeast margin of the field where clockwise rotations as great as 90° have been postulated at the northwest end of the Las Vegas Valley shear zone. Any clockwise rotation at the northwest end of the Las Vegas Valley shear zone must predate 13.25 Ma, and thus dextral shear within this part of the Walker Lane belt was not synchronous or connected across the entire margin of the field.

Small counterclockwise vertical-axis rotation as suggested for the central SWNVF block may be a regional feature of post middle Miocene deformation in the western Great Basin. This rotation may reflect a province-wide strain gradient in which northern parts of the province experienced greater cumulative extension as they were translated away from the Colorado Plateau.

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